

Still Modernism: Photography, Literature, Film

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Resisting Arrest

Imagining Motion in Film and Photography

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Abstract and Keywords

This chapter examines how early filmmakers had to invent what motion looked like on screen, imagining it as distinct from stillness, legibility, or clarity. The images of motion in early film are blurred and impressionistic—ocean waves, clouds of dust, puffs of steam and smoke—which render motion itself a kind of obscurity and reveal how film is itself an ephemeral medium of dust and smoke. The precursor to film's absent materiality is found in photography's own representation of motion as blur in Etienne-Jules Marey's strange late nineteenth-century photographs of smoke fillets and the movements of air. These images, lesser known than his other motion studies, reveal how film casts back to its still antecedent to imagine motion in blurred terms of smoke and dust, even as it resists photographic arrest.

Keywords: early film, motion, speed, smoke, F. T. Marinetti, Lumière brothers, Cecil Hepworth, Etienne-Jules Marey

But in film the still image is in vain, like the foreboding of a car
breakdown, like watching out for death.

—Agnès Varda, "On Photography and Cinema"

I saw trees growing and changing like puffs of vapour, now brown, now
green; they grew, spread, shivered, and passed away.

—H. G. Wells, *The Time Machine*

One of the most important factors in obtaining a good negative is a pure atmosphere. Particles of dust, floating in the air, when illuminated by the sun, form a sort of luminous haze, which interferes with the clearness of the photographic field.

—Etienne-Jules Marey, *Movement*

Motion pictures' rejection of stasis is part of film's history, nomenclature, and production and constructs a good part of its lore as well. Film historians have long elaborated on how the first demonstrations of the Lumière brothers' films in 1895 dramatized the fundamental shift from stasis to animation, which informs the central tenet of film's claim to medium specificity through an aesthetics of movement.¹ Early film screenings began by projecting an initial still image, which would resolve into motion after an anticipation-filled delay.² The original freeze-frame is a point of stoppage set up only to be overcome and relevant (so the story goes), mainly because it augurs its opposite and its undoing in the name of technological advancement. The still image points away from itself, **(p.19)** drawing attention to what is about to happen and to the apparatus as a machine of and about motion by way of its finite suspension. Not only does motion write itself into film as a visual rupture on the level of apparatus and screen, but also, as the Lumières's well-known film *Arrival of a Train at La Ciotat* (1895) demonstrates, motion (or, as Laura Marcus puts it, "extreme motion") becomes the very theme and subject of the first motion pictures.³ The apocryphal story surrounding this short film—that audience members, terrified by the sight of the onrushing train, fled the scene, fearing they would be run over—is an origin myth that plays up the fear and exhilaration of speed and acts to bind the two technologies of motion (the railway and cinema), which both alter fixed notions of time and space. The fear of the moving train represents a fear augured by the new motion technology: that it will not stop; that it would not actually arrive anywhere. This terror of not stopping is the flip side of the initial freeze-frame's manufacturing of the expectation of motion, which might also be understood as the fear of not moving. Strangely, then, both motion and stillness are presented as inverted problems in early film.

This chapter begins with the Lumières's *Arrival of a Train* for the ways in which it articulates how the new medium of film describes motion as unstoppable. In exceeding the two-dimensional limits of the still frame and fixed image, early film invented not only motion pictures (a technological innovation) but also what motion looks like on screen (an act of imagination). Motion emerges in film, initially, as a counterfactual—not stillness. As *Arrival of a Train* makes clear, early film's devotion to the representation of motion frequently meant the explicit avoidance of stillness and of still subjects. Not surprisingly, the cinematic ideology of kinesis coalesces in the context of other modern systems of acceleration and transport. The railway, the streetcar, the automobile, and the airplane offer new modes of spectatorship and new thresholds for the experience

of speed that pivot on their affiliation with novelty and spectacle and underpin modernism's forward-moving, accelerative culture, yoking speed to ideologies and technologies of progress. Following Wolfgang Schivelbusch's work on train travel and perception in the nineteenth century, Lynne Kirby, for instance, posits a perceptual overlap between the railroad and the cinema, arguing that the train provides the "prototypical experience of looking at a framed, moving image" and is the "mechanical double for the cinema."⁴ She likens film's illusion of movement to the railway's own negotiation of immobility and mobility: "a passenger's being at once immobile and in rapid transit, lulled to sleep and yet capable of being shocked awake."⁵ What seems to be a repudiation of forms of stoppage in favor of expressions of speed, which is evident in early film's immersive (p.20) embrace of technologies of velocity, shores up a rhetoric of motion that cultivates traction precisely by rejecting stillness as its undesired opposite. Kicking up clouds of dust and obscuring the screen, filmed trains rush down their tracks and motor cars speed by, capitalizing on a refusal to stop that speaks to a cultural moment infused with the desire for ever-increasing modes and methods of mobility.

"We intend to exalt movement and aggression, feverish insomnia, the racer's stride, the mortal leap, the slap and the punch", F. T. Marinetti writes in the "Manifesto of Futurism" (1909).⁶ The aim, he says, is to transcend earlier literary models that valued "contemplative stillness, ecstasy and sleep." But despite Marinetti's declarations, in this context of increasingly prevalent technologies and ideologies of motion, stillness is not actually overtaken by kinesis, but, I argue, lingers as a perceived threat or source of fear. Indeed, the peregrinations and manipulations that are involved in film's avoidance of stillness shade into the ridiculous in a corpus of early British film comedies that are all about cars that cannot or will not stop, as some of their titles suggest: *How to Stop a Motor Car* (Percy Stowe, 1902) or *How It Feels to Be Run Over* (Cecil Hepworth, 1900). I read these films in the context of Futurism's own dogmatic embrace of speed to show how velocity is made over as something ridiculous—something that produces a preposterous and perverse narrative of relentless motion that comports with the blurring and blurred effects of an accelerative culture.

If early film represents motion by resisting stillness, what does motion in the absence of stasis end up looking like? Film's uniqueness is that it could represent a moving image sharply, keeping pace with its moving subject. However, I show that early film nonetheless elaborates on the still image's use of blur and abstraction as a means to imagine motion in terms inherited from photography. Blurriness is speed's opaque symptom, developed by still photography and turned into a visual and material trope of film. Although films could move and represent motion, they did so by obscuring it, reinforcing motion's abstraction or invisibility and film's own ephemerality. Film's blurred images of motion force us to look past what is visible, to look through the screen

and reconstruct the source of motion. Motion blur is a feature of still photography, which marks its radical remove from continuity. It indicates movement that could not be apprehended, that eludes the speed of the shutter. The long exposure times of early photographic images meant that a still photograph of any moving object was bound to be blurred, which faster shutter times attempted to remediate. Film does not necessarily deal in motion blur, but in blurriness writ large, transforming the source of the opacity from a photographic incompatibility, which marks **(p.21)** the tension between stillness and motion, to a filmic representation, which turns motion into a visible quality of the ephemeral. The visual narrative of endless motion in some of these early films is represented by a series of overlapping tropes that are like *sfumato*—dust, smoke, steam. These hazy particulates mark film's attempts to render motion itself visible as a blurred abstraction, challenging the solidity or clarity of stillness. The dust and smoke in early films typically emerge from other contemporary technologies of motion—steam trains and motor cars—that seek to disavow a pedestrian, or slowed, pace in favor of increased modes of acceleration. The iconic filmed image of a moving train emerging from clouds of steam contributes to how the newer motion technology—the motor car—made its way into a filmic discourse of motion that also sets up smoke screens to blot out stillness. The clouds of smoke that are symptoms of speed accrue on film's surface and are put to the task of obscuring stillness itself as a means of depicting motion. To put it succinctly, we see motion by not seeing clearly.

Not much is written about photographic blur, perhaps because since the invention of photography, the effort has been to eliminate blur from photographs and to sharpen photography's temporal precision, expunging the unwanted visibility of duration or the mark of time's passage. Clear photographs, achieved through technological advances, are valorized in most conventional histories of photography, which follow on the mandates of modernism and aim for a particular artistic status to attach to the photographic object. Any discussions of blur usually have less to do with motion than with pictorial photography's imitation of painterly techniques in the nineteenth and early twentieth centuries. Although soft focus might have bestowed on photography the status of fine art at the end of the nineteenth century, the blur that was criticized even in the earliest photographs had to do with what Larry Scharf calls "anonymous smears or ghostly vestiges", which were the reminders of something that had been in motion during the long exposure of the image, but that could not be seen as anything but a shadowy form.⁷ The slow shutter speeds and long exposure times of early photographs meant that blur attached to the edges of things, despite all attempts at stillness and precision. The invention of a more sensitive ground and faster shutter speeds expunged unintentional blur from the photographic image, only to be brought back, as we have seen, in a cloud of dust, after the invention of film. In other words, despite its ability to represent motion by motion technology, film's *sfumato* inherits photography's blur. Motion's blur, no longer a

problem of the still camera, moves from articulating the limits of the visual field to the volumetric clouds of the atmosphere.

(p.22) Smoke Screens

In the Lumières's famous film, a train pulls into a station, emerging from the perspectival vanishing point on the upper right-hand side of the screen and moving diagonally from background to foreground. Carrying on beyond the fixed camera, which is set up on the platform, the moving train's trajectory exceeds both the frame and the limits of perspective, insisting on spatial continuity. As the train's engine disappears off-screen, its new foregrounded vanishing point is unstable, a result of being too close, rather than a mathematical product of the traditional perspectival planes of distance. In an 1896 essay called "The Cinematograph", O. Winter observes that the Lumières's camera exercises the "monstrous privilege of enlarging the foreground beyond recognition", rendering the train unseeable.⁸ The train's movement from background to foreground not only makes the locomotive object loom ever larger but also makes it look like it is actually accelerating as it approaches the fixed camera.⁹ The train and the scale of motion thus exceed the frame (Rudolf Arnheim calls this "a dynamic dilation towards the margins of the screen"), which contributes to an overall sense that the enormous train is fast approaching, when it is, in fact, stopping.¹⁰ The amplification of motion and space by perspectival illusion privileges speed and not stopping as the dominant visual tropes of this short film.

The speed of the train (actual or perceived) produces a series of blurred images, setting up a particular cinematic inability to see that is conditioned by motion. The frame's vanishing point at the top of the screen, which is the origin of the train, is completely obscured by the steam emanating from the engine. The train is in view as the film begins, rendered visible by motion itself. However, its rapid pace clouds everything in its wake: the train emerges from deep space, which is obscured by motion's blur; layers of white and gray steam become the residue of mechanical speed and a figure for the illusion of film itself. The Russian critic Maxim Gorky acknowledges cinema's sidestepping of linear perspectival origins when he discusses the impact of witnessing motion in *A Street in Paris*, which he saw in a program of Lumière films (including *Arrival of a Train*) at a fair in Russia in 1896.¹¹ Emphasizing the cinema's volume, he describes the disorienting effects of "carriages coming from somewhere in the perspective of the picture", which "move straight at you", penetrating the darkness of the audience's space in the theater. The dislocated screen yields to people appearing "somewhere from afar", growing larger as they approach. The street scene he describes, which "teems with life", approaches "the edge of the screen" and "vanishes somewhere beyond it." Gorky's carriages have no **(p.23)** perspectival origin—"coming from somewhere", they simply move into the space of the

screen and beyond. Their motion, which is amplified by proximity, then suddenly disappears into deep space.

In Gorky's description of cinema's three-dimensionality, motion extends beyond the edge of the screen and disappears from sight. In *Arrival of a Train*, the train's engine continues past the fixed camera to an unseen destination off-screen. In other words, we do not see the engine stop; we only see its disappearance from the frame. As the engine passes the camera and begins to slow, the people waiting on the platform adopt its motion. Drawn to the black hole of the edge of the screen, a conductor runs down the tracks at the same speed as the train and also disappears from view. Matching the speed of the human body to the speed of the machine is characteristic of many train movies, when people run to catch up with departing trains. Here, however, the conductor is running to catch up to a train that is slowing down. Even after the train has stopped, the passengers continue its forward-moving diagonal trajectory, walking toward the front of the train and the front of the screen. Passengers embark and alight, and the platform becomes a site for the flurry of travel. Their rapid motions work, thus, to conceal the stillness of the train. Although the train stops, it does not terminate, since the carriages extend beyond the pictorial frame in the foreground and are obscured by the steam in the background.

The train itself has neither beginning nor end, neither origin nor terminus. It emerges from nowhere—a puff of smoke at the mouth of the vanishing point's cave—and arrives nowhere—off-screen, a beyond. This transgressive motion disrupts the fixity of the still vantage point of the camera, as is the case, too, in a corpus of early train films by Edison, such as *Empire State Express* (1896), *Black Diamond Express* (1896), *Philadelphia Express* (1897), *Freight Train* (1898), *Overland Express Arriving at Helena, Mont.* (1900), and *Arrival of McKinley's Funeral Train* (1901), where the train emerges from a background clouded by smoke and crosses diagonally on curving tracks past the fixed camera, traveling onward somewhere beyond the pictorial frame. This unseeable beyond, like the film itself, pushes past stillness toward continuity and the desire to express movement that does not have to stop. Stillness, if it exists at all, does so outside the frame, which becomes a form of negation—a way of obscuring sight.¹²

Shortly after *Arrival of a Train* begins, the entire left-hand side of the screen is filled with the body of the moving train, whose shiny surface intersects the plane of the cinematic screen from top to bottom. Many of Edison's train films share this same relationship of train and frame, with one side of the screen entirely filled by the train cars that pass through the intersection of moving train and motion picture. The replacement of (p.24) screen by moving train is further animated in Edison's *Philadelphia Express*, *Jersey Central* (1897) because of its low vantage point. The low point of view flattens the depth of field, emphasizing the verticality and flatness of the screen against motion's unfolding. By isolating

the moving image from its surroundings (from points of origin or arrival), the train/screen turns motion itself into an abstract visual form. And what motion looks like is a shiny surface blurred by shadows and light: a film screen.

The black-and-white color scale of these early films underpins their crosshatching of motion with abstraction and blurriness, the sense of rushing headlong into space. Gray is the color of blur, and Gorky comments on the grayness of the Lumière's films in his account:

Everything there—the earth, the trees, the people, the water and the air—is dipped in monotonous grey. Grey rays of the sun across the grey sky, grey eyes in grey faces, and the leaves of the trees are ashen grey. It is not life but its shadow, not motion but its soundless spectre. . . . It is terrifying to see this grey movement of grey shadows, noiseless and silent.¹³

Gorky's fear of mute grayness has to do with its vagueness and imprecision, which dulls the spectator's consciousness: "You are forgetting where you are." He understands the gray imprecision of film in terms of a failed realism, borrowing the silent shadowy illusions from Plato's cave. The shadowiness of film is figured in *The Arrival of a Train* by the train's smoke—the great puffs of steam that make the perspectival position of the moving train unknowable. The analogy thus drawn by film is between its motion per se and shadow or smoke. Identified as a technology of motion, film is a medium of speed, smoke, dust, and obscurity—the stuff of fast-moving trains and accelerating motor cars.

The obscured vision resulting from the film's embrace of white clouds of steam is a product of speed. Another description of the film from 1896 concurs, pulling together an account of the medium of film in terms of smoke, speed, and the accident:

In the distance there is some smoke, then the engine of the express is seen, and in a few seconds the train rushes in so quickly that, in common with most of the people in the front rows of the stalls, I shift uneasily in my seat and think of railway accidents.¹⁴

The smoke obscures the origins of motion in stasis, as the fast-moving train glosses over its vulnerability to stoppage, the accident, and the frame. The fear of catastrophe that haunts this early encounter with film (**p.25**) is underpinned by the unexplained and unseeable origin of the motion, which is clouded by smoke and a distance traversed too quickly. The uneasiness of the scene turns on the anxiety that motion might suddenly cease. The anticipation writ into these early screenings is thus twofold and contradictory: the audience anticipates motion from the still initial image and, subsequently, anticipates unwanted the possibility of stillness's return. The "atmosphere of expectation" cultivated by these filmic spectacles hinges on the instant when motion bursts forth into space and the possibility that it will cease. Whereas the filmgoer witnesses the former

transformation from still image to moving picture, from stillness to speed, the latter transformation, the accident (the stopped train, the original stillness), is obscured by screen and smoke, which become markers—visual residue—of motion cast against and beyond stillness.

Overcoming Stillness

Classical film theory (André Bazin, Siegfried Kracauer, Walter Benjamin) builds from the notion embedded in these early train films that forms of stillness are something to be overtaken or obscured—indeed, to be triumphed over by cinema. Accordingly, the conventional understanding of film is that cinema essentially derives from the ontology of the photographic image and, by the manipulation of stills, can in fact act like photography—only better, as it can also move. Cinema's great achievement is kinesis: composed of a sequence of individual photographic frames (but not individual photographs), the technology works by turning the still image into something that moves, erasing the frame and producing an illusion of continuity. According to this model, cinema both solves and assimilates photographic elements of fixity and stillness—and in the end, negates the difference between still photography and motion film from the point of view and to the advantage of cinema, a more expansive medium. In other words, motion pictures do away with photographic stillness (effacing its limitation) while appearing to enfold and incorporate the photograph. Early commentators on the medium declared the immanent death of photography at the hands of the newer motion technology—for what good is the still, frozen image once you have the ability to make it move? Georges Dmenÿ, for instance, who assisted Etienne-Jules Marey in his nineteenth-century photographic studies of motion, declared in 1892, "The future will see the replacement of motionless photographs, frozen in their frames, with animated portraits that can be brought to life with the turn of a handle."¹⁵ Dmenÿ's prediction did not come true (the still photograph clearly **(p.26)** stayed, taking up its position both within and against animated forms), but his comments nonetheless reveal an important understanding of photography in modernism that has to do with the medium's limitations—the image frozen and framed—which he deems a lack, worthy of replacement. Dmenÿ's vision for the future speaks to an underlying premise or hope, which marks much of the writing about film in the early twentieth century, that film will overshadow (overrun, overtake) still photography.

Like Dmenÿ, commentators on early film sought to bracket the newer motion technology from photographic stillness—to cover over the photographic referent beneath the film strip and shift attention from the material base of film to its surface or screened effect. Their insistence was almost always on film's ability to move, its embrace of a dynamism, continuity, and immediacy that trumped photographic stasis and relegated photography to either the status of anachronism or a passing stage on the way to cinematic victory. And so, in 1920, Louis Delluc called for cinematic medium specificity, under the auspices of what he named *photogénie*, which has everything to do with motion and immateriality:

“Photography is not *photogénie*. . . . No more photography! Cinema! All the resources of photography and the ingenuity of those who revolutionized it will come to serve—but what a sumptuous servitude and devotion!—the fever, the sagacity, the rhythm of cinema.”¹⁶ Delluc puts into play a strange analogy of conquest, based on a formal argument that cinema both replaces and incorporates photography, which is cast into the position of servitude. Photographic meaning is divorced from its original source and made over as necessarily cinematic. Interestingly, Delluc’s attempts to define what he calls “the purest expression of cinema” produce a mixed result despite the desire for uniqueness. Film is (or was), after all, photochemical. *Photogénie* thus has to do with the interweaving of cinema and photography (“l’accord du cinéma et de la photographie”), which sees the photographic image put to cinematic use, stripped of its material form.

Delluc is by no means alone in identifying motion as the ontological base of film—by the 1920s, stasis per se seems to have become uncinematic and was branded as antimodern.¹⁷ Elaborating on Delluc’s notion of *photogénie* and its blend of medium specificity and medium superiority, in “On Certain Characteristics of *Photogénie*”, Jean Epstein writes, “Every art builds its forbidden city, its own exclusive domain, autonomous, specific, and hostile to anything that does not belong. Astonishing to relate, literature must first be literary; the theater, theatrical; painting, pictorial; and the cinema, cinematic.”¹⁸ Epstein does not mention photography in his list of forms of artistic decorum but distinguishes cinema from literature, theater, and painting, staking its artistic claim by association with (p.27) received forms of art as much as originality. Accordingly, he does not offer the status of art to photography, reserving this aesthetic appeal for the newer technology of film.¹⁹ Later in the essay, Epstein tacitly acknowledges photography as the very thing (immobility) that cinema (motion) corrects: “the mechanism of cinema constructs movement by multiplying successive stoppages of celluloid exposed to a ray of light, thus creating mobility through immobility, decisively demonstrating how correct the false reasoning of Zeno of Elea was.”²⁰ Epstein solves Zeno’s paradox of the arrow by way of cinema, which undoes the opposition of stillness and motion. Zeno suggests that at any given moment, an arrow in flight is, in fact, at rest, occupying a stage along the way; continuous movement is subtended by discontinuity, which means, according to his “false reasoning”, that there is no such thing as movement. Epstein counters that the cinema successfully produces motion from stillness—stillness operates at the service of cinematic motion, whose illusion erases the gaps and discontinuities. Whereas photography fixed the moving object (what William Henry Fox Talbot referred to as moving shadows in *The Pencil of Nature*), film, in turn, repairs stasis, restoring motion by application of the photochemically fixed image. For Epstein, film’s unique quality has everything to do with this synthesis of stasis and motion, which claims to render immobility invisible.

The cinematic drive to obscure photography's stillness not only solves Zeno's paradox by rendering stillness as motion, but also is envisioned as a dual triumph over both photography and death. Here, we read an account of motion pictures from 1895:

It is life itself, it is movement taken from life. . . . Photography no longer records stillness. It perpetuates the image of movement. When such gadgets are in the hands of the public, when anyone can photograph the ones who are dear to them, not just in their immobile form, but with movement, action, familiar gestures, and the words out of their mouths, death will no longer be absolute, final.²¹

The cinematic refusal of stasis is deemed a refusal of the finitude of death. It performs a guarantee of continuity or presence in the face of mortality and puts in place a historic duality that would find, in the limits of the still photograph, a fixation on a moment marked by the past and, in the ephemerality of moving film, the representation of the flow of life.

The life-and-death opposition of film and photography underwrites André Bazin's well-known essay, "The Ontology of the Photographic Image", in which he sets up film as a broader representational solution that **(p.28)** emerges from photography—film is like photography, but capable of so much more; indeed, it is capable of rescuing life from death:

Film is no longer content to preserve the object, enshrouded as it were in an instant, as the bodies of insects are preserved intact, out of the distant past, in amber. The film delivers baroque art from its convulsive catalepsy. Now, for the first time, the image of things is likewise the image of their duration, change mummified as it were.²²

Whereas Delluc saw film as a conqueror, Bazin sees it as a savior, an anodyne to the disease of baroque art, which, as he discusses earlier in the essay, suffers from a "tortured immobility" despite its quest for motion. Bazin is interested in positioning cinema and photography as technological solutions to centuries-old representational concerns, constructing a sweeping view of art history. However, the phrase "convulsive catalepsy" seems to describe the relationship and historical proximity of the invention of film and photography, rather than their joint distance and appeal to the historical concerns of baroque art. Cinematic continuity smooths over the rhythmic jerks and paralysis of the snapshot or instantaneous photograph. The filmic archive allows for the duration of the image, for it to unfold in time, whereas photographic fixity freezes and hesitates at the moment of preservation. To use another phrase from Bazin, the photograph "embalms time"—and forecloses on the future. Bazin's teleology is one of defiance, with cinema representing a final triumph over time and over death as it unfolds in the present tense. You can hear the same triumphant tone

in Christian Metz's important essay, "Photography and Fetish", when he writes that "film is less a succession of photographs than, to a large extent, a destruction of the photograph, or more exactly of the photograph's power and action."²³ Film—an ephemeral projection—unfolds in the present tense, occupying a shared temporality with the spectator. It thus trumps the "power" of the photograph, which apprehends only finitude, immobility, and death.

Speed's Thrill

In reading these descriptions of filmic wonder and triumph, it is difficult to know how seriously to take the metaphors that unequivocally align photography with death and film with life, which ultimately paper over a fear of death by dismissing stillness.²⁴ The fear of death is figured by its opposite—a single-minded celebration of vivacity by way of speed—and **(p.29)** propelled by technologies of kinesis and movement, which are taken up uncritically as hyperbolic signs of immortality and triumph. These are the terms that provide the framework for thinking about modernism's hyperbolic ode to speed, F. T. Marinetti's first Futurist manifesto, published on the front page of *Le Figaro* in 1909. This text, in which Marinetti famously celebrates the figure of the speeding automobile and the "beauty of speed", reveals the particular ways in which the Futurists leveraged technology to try to render stillness moot in the early part of the century.²⁵ Marinetti describes an intoxicated drive he took four months previously, which resulted in a car crash, an unwanted stilling that provokes the driver's desire to resist the state of arrest and to speed away from the site of the accident; in other words, he uses speed to obscure the origins of modernism as still. Marinetti insists we understand the crash at the center of the story in terms of what will become the signatures of a Futurist ethics: action, departure, disappearance, newness, and the future. The crash, which is the founding event of Futurism, is purposed as an emblem not of stillness, but, as Jeffrey Schnapp has noted, of "aggressive movement", celebrated in Marinetti's text for its association with speed, a new form of violence, and the thrill of danger.²⁶

Marinetti's manifesto is frequently named as a critical example of the linking of speed with pleasure.²⁷ As Enda Duffy explains in *The Speed Handbook*, speed's pleasure is twofold, deriving at once from the experience of moving very fast and at the same time from the "sensation of controlling that movement", which allowed people to "feel modernity in their bones."²⁸ James Joyce writes in "After the Race", the short story in *Dubliners* that begins with cars "scudding in towards Dublin" as they near the end of the Gordon-Bennett automobile race, "Rapid motion through space elates one."²⁹ In a later essay, "The New Religion-Morality of Speed", Marinetti describes the "inebriation of great speeds in cars" as "simply the joy of feeling oneself merged with the only *divinity*."³⁰ If the experience of speed, here understood as velocity shot through with male fantasies of power, is linked inextricably to pleasure, then the stilling of motion, the stopping short that causes Marinetti to roll into the "maternal ditch", carries

with it a particular stripe of *displeasure*, marked by a dual lack of speed and lack of control. Stoppage, in this context, reinforces an understanding that the desire to control speed is predicated on a fiction. Marinetti's unwanted return to stillness and the womb is an inevitable result of speed's overdetermined pleasure.

C. S. Rolls, the founder of the British car company Rolls-Royce, explains the problem of stopping this way, when protesting the campaign for a new speed limit in Britain in 1903: "Our hereditary instincts are shocked at (p.30) seeing anything on the road faster than a horse, but as our senses become educated we shall recognize the fact that speed itself is not dangerous but the inability to stop is dangerous."³¹ According to Rolls, there is nothing dangerous about the *experience* of speed, which he understands simply as the apprehension of a different scale of motion—a quickening of the eye that he says can be learned.³² Rolls said later in an interview with *The Guardian*, that "the idea of having a specific speed-limit at all is wrong. . . . The speed-limit ought to be utterly abolished, and there should be substituted a strenuous scale of punishments for accidents occasioned through deliberate or undeliberate fault of the driver." In other words, the accident (the stoppage) ought to be punished, not speed per se.

Rolls's comments on speed offer a counterpoint to the well-known warnings issued by critics of modernity like Georg Simmel or Walter Benjamin, who both caution against the effect on the untrained eye of the shifting and chaotic images of modern everyday life, elaborating on a lexicon of trauma and speed. Whereas Rolls is confident that the senses can learn and eventually understand the coordinates of faster speeds, in "The Metropolis and Mental Life" Simmel describes the suffering of the "metropolitan type of man", whose overstimulated nerves are driven to numbness under the guise of self-preservation. The urban dweller cultivates a "blasé attitude" to counteract "the rapidly changing and closely compressed contrasting stimulation of the nerves."³³ Benjamin draws heavily from a Freudian model and assumes that the eye, conditioned by the onslaught of proliferating images in the city, becomes an organ of protection (resulting in a kind of blindness) rather than a means of sight or knowledge.³⁴ Rolls, who in 1903 was deeply invested in turning speed itself into a consumer product, does not equate acceleration's relationship to the eye with traumatic blindness or with the protection against stimuli, as Benjamin would have it. For Rolls, the shock of the automobile was simply a matter of people not being used to seeing fast-moving vehicles, rather than a fundamental inability to parry the modern sensation of speed. He declares that the experience of speed is safe for consumers ("speed itself is not dangerous"). The danger lurks in the physical properties of speed, like entropy and momentum, that run counter to stopping. In other words, the danger is in trying to stop, not in trying to go quickly. For Rolls, the perils of stopping are manifest and made punishable by the car accident—a stoppage that is unexpected and unwanted; a stilling of motion in spite of itself; a rupture of both physics and the status quo. Rolls implicitly

acknowledges that the technology of the motor car invokes what Paul Virilio has called “its own negativity”: that is, the car crash emerges alongside the car. Once you have a technology that is devoted to acceleration, you also have a technology that will crash. “To **(p.31)** invent the family automobile”, writes Virilio, “is to produce the *pile-up* on the highway.”³⁵

The car crash, for Marinetti as for Rolls, elaborates a specific problem in modernism and theories of modernity that has to do with the difficulty (and necessity) of stopping in a context that is otherwise defined by the rhetoric and technology of speed. Marinetti’s crash is caused by two cyclists, who appear suddenly in the way when Marinetti spins his car around. He writes that these two characters, “wobbling like two lines of reasoning, equally persuasive and yet contradictory”, present as a “stupid argument” against his own velocity, “cutting me off, as if trying to prove me wrong.”³⁶ The presence of the two cyclists on the road, who disappear from the driver’s vantage as suddenly and quietly as they enter the picture, is repurposed as a rhetorical device—two logical or sober counterpoints to the actions of a spinning car and its drunk driver. Stillness or the need to stop—to hesitate in the face of something in the way—thus lurks as a dialectical threat behind Marinetti’s discourse of relentless and restless motion. Although not themselves still, the cyclists nonetheless are impediments. Lutz Koepnick remarks that it is the very slowness of the cyclists that causes the driver to veer into a ditch.³⁷ They, and their blocking of the road, become anachronisms, whereas speed speaks in the celebratory tense of the future.

Boredom’s Roadblock

On encountering the obstacles in the way, the narrator of Marinetti’s escapade exclaims, “What a bore! Damn!”³⁸ But why invoke boredom? Shock, surprise, fear, relief, or anger, perhaps, but Marinetti’s *ennui* makes little sense except as an annotation that he appended to bring the emotion’s leaden temporality to bear on his disgust at having to stop. Boredom is the affect of stillness. Marinetti is not describing the sort of prolonged or quotidian event that we would normally associate with the boring; nonetheless, he leverages boredom’s lag as a way to dismiss stasis on account of its own affective qualities, which dismantle speed’s pleasure.³⁹ The deeply strange association of the temporality of boredom with the suddenness of the event has not to do with the crash, which happens quickly, but with the very fact of stopping short at all, which is a return to a state of latency in the maternal ditch. The boring is established by the impediments in the road—the cyclists, who preclude motion and represent an outmoded means of attaining speed.⁴⁰ Although boredom typically describes the emotion of enduring a prolonged period of time in which nothing happens, here it **(p.32)** becomes the event itself, the threat of stoppage or of something getting in the way of speed.⁴¹ This threat is cast in terms of the maternal, whose stagnant ditch represents an unwanted return to a primordial enclosed origin.

Later, in his essay “The New Religion-Morality of Speed” from 1916, Marinetti reiterates his opposition to things being in the way when he writes,

Velocity = disdain for obstacles, desire for the new and unexplored.

Slowness = arrest, ecstasy, immobile adoration of obstacles, idealization of tiredness and repose, pessimism about the unexplored.⁴²

Speed, under this rubric, attaches to pleasure and the unencumbered pursuit of the new, whereas stoppage becomes its reverse, partaking in paralysis and blockage. Stillness represents the failure of speed and, consequently, the failure of pleasure. Marinetti’s exclamation (“What a bore!”) serves to summon boredom’s *sluggishness* (to borrow Sianne Ngai’s description); the exclamation elicits boredom’s “deficiency of affect”, which then shapes the crash’s own deficient ethical and emotional impact.⁴³ Stopping short, the car separates from the pleasures derived from velocity and hews instead to the dull tedium enacted by stillness. For Marinetti, the driver, the sluggishness of boredom gives way to the impatience of disgust, which emanates as much from the vehicle’s stoppage as it does from the stagnant mud and muck (“bracing slime”) of the maternal ditch from which the driver emerges following the crash, filthy and ready to declare “a new form of beauty: the beauty of speed.”⁴⁴ The celebration of the aesthetic category of speed gathers its own momentum from the imperative inherent in disgust to push away the monotony of its static object of aversion. Disgust brings with it the rapidity of recoil, which produces an automatic and impulsive motion that refuses stoppage and is associated with speed.⁴⁵

The “sin” of slowness or stasis derives from its visual epistemology, which cultivates an ethics of prolonged looking based on return (what Marinetti disparages as “the already seen”) and the sustained gaze.⁴⁶ Remember, Marinetti does not return his attention to the original impediment—the cyclists in the road seem to disappear—but dispenses of them through narrative continuity, or the desire, simply, to move on, to move ahead, to not look back.⁴⁷ Stillness, which under Marinetti’s strictures is gendered female, inhabits the prolonged temporality of boredom and puts up with obstacles (wobbling cyclists) and stagnant, unhygienic ditches, which run counter to speed’s immaterial sensation.

In a review essay from 1918 called “The Movie Novel”, Virginia Woolf parodies an impulse like Marinetti’s to leverage speed to mitigate the static **(p.33)** temporality of boredom. In her essay, which echoes (unwittingly, perhaps) Marinetti’s dual categorization of velocity and stillness, Woolf embeds a critique of cinematic speed alongside a review of Compton Mackenzie’s novel *The Early Life and Adventures of Sylvia Scarlett*. She writes,

Although Mr Mackenzie can see [his characters] once he can never see them twice, and, as in a cinema, one picture must follow another without stopping, for if it stopped and we had to look at it we should be bored. Now it is a strange thing that no one has yet been seen to leave a cinema in tears. The cab horse bolts down Haverstock Hill and we think it a good joke; the cyclist runs over a hen, knocks an old woman into the gutter, and has a hose turned upon him. But we never care whether he is wet or hurt or dead. So it is with Sylvia Scarlett and her troupe. Up they get and off they go, and as for minding what becomes of them, all we hope is that they will, if possible, do something funnier next time. No, it is not a book of adventures; it is a book of cinema.⁴⁸

Disparaged as a “book of cinema”, MacKenzie’s novel presents, for Woolf, as a failure because of the relentless pace of its characters and their actions: “up they get and off they go.” Comparing the ceaseless motion of MacKenzie’s characters to the technology of the cinema, Woolf claims that they ultimately refuse identification.⁴⁹ The fast-moving body becomes inhuman, devolving instead into a mobile collision site. The cyclist does not get hurt as he careens about; or, even worse, Woolf says, if he does injure himself or those around him, “we never care whether he is wet or hurt or dead.” The cyclist’s devotion to movement at all costs and despite impediments means that we do not recognize the characters’ suffering or possible death—and it is this inhumanity that Woolf identifies as both the source of the joke and precisely the problem with cinema, understood here as an art devoted singularly to speed. In the context of the moving series, both death and the recognition of death are indefinitely postponed. Woolf thus offers a concise indictment of cinema on ontological grounds, linking its affiliation with motion and speed—its desire not to linger or stop—with its affective inadequacy in the face of death.

Although Woolf does not pursue a full-fledged argument about cinematic motion conceived against or in contrast to photographic stillness, she nonetheless suggests the terms by which we might understand the pursuit of stillness as a way *not* to defy death. The excess of movement that underpins the cinematic form is symptomatic of its flinch or start from death, which is concealed by linking motion to humor. Under the mandate of kinesis, the comic body becomes dehumanized as an automatic motion machine. So, too, MacKenzie’s characters, as Woolf describes them in her (p.34) review, avoid a narrative of death or finality by way of automated humor—the privileged narrative is of one-upmanship; “the hope”, as Woolf says, is “that they will, if possible, do something funnier next time.” She complains, “You can scarcely open the book anywhere without finding a cab bursting down Haverstock Hill with an eloping couple inside it, or a baboon escaping from Earl’s Court Exhibition, or an actor dropping dead, or a curtain going up, or a landlady being funny.”⁵⁰ As has been noted by Laura Marcus, Woolf’s criticism resonates with early accounts of film that caution about the medium’s proximity to what W. T. Stead calls “life caught

in the act of living.”⁵¹ In a review of cinema from 1914, he writes, “The rapid and constant succession of moving pictures leaves no time for reflection. You see life as from the window of an express train. You have not even opportunity to recollect the impression of the scene.”⁵² The problem he identifies is with the very speed of film, which produces an excessive mimesis in which spatial detail accumulates relentlessly, producing an aesthetic of immediacy and invisibility on which comedy plays.

Woolf critiques MacKenzie for refusing to allow anything to properly get in the way of the moving body. A hen is summarily run over, an old woman thrown into a gutter, and the cyclist himself soaked, but nothing comes to a stop. Although Woolf’s claim that “no one has yet been seen to leave a cinema in tears” ignores the sentimentality of many early films, which elevate pathos over action (such as the films of D. W. Griffith, for instance), her exaggerated argument about film’s inability to stop, even for death, serves her main point that the contemporary unchecked celebration of speed devolves into the ridiculous.⁵³ In his essay on laughter, Henri Bergson also affiliates boredom and incessant motion with an affectless humor. The comic effect, produced by movement or momentum, demands the “momentary anesthesia of the heart”, an emptying out of emotion, that nonetheless remedies boredom’s temporal drag toward stillness.⁵⁴

The fear that underpins MacKenzie’s relentless continuity and the cinematic injunction that “one picture follow another without stopping” runs parallel to Marinetti’s own base fear that the cessation of motion would produce boredom; that an impediment recognized as such would produce an unwanted stillness and accommodate the stoppage required by death. Whereas Marinetti’s overdetermined boredom gives way to an alternative passion of disgust, the flip side of boredom in Woolf’s case is laughter. Rather than producing tears, the novel under review summons the kind of humor or lack of identification that Woolf associates with film’s motion. Both the disgusting (Marinetti’s maternal ditch) and the risible (Woolf’s wet cyclist) turn away from the temporal threat posed by boredom’s lethargy, the former co-opting impatience by way of recoil’s acceleration and **(p.35)** the latter summoning the ephemerality and spontaneity of the present moment through mechanical imitation.⁵⁵ The ceaselessness of motion that Woolf criticizes harbors a cinematic joke, which pivots on interruption and speed. Each action takes over from the previous one and dismisses the purchase of the impediment and of stoppage. Thus, at the center of a modernist understanding of and dedication to speed lurks the twinned threat of boredom and stillness.

No Stopping

As Woolf implies by her comparison of MacKenzie’s accelerative novel with cinema, many early films explicitly avoided stillness and did so by doggedly depicting motion that keeps going at all costs, self-consciously refusing the limitations of the photographic frame. The out-of-control and uncontrollable cars

depicted on screen refuse to accommodate motion's antithesis, a refusal that underpins the central ideology of Futurism. But ceaseless motion in the absence of stillness or stoppage is as perverse, or as preposterous, as an instantaneous photograph that catches a body in flight, forever suspended, its action incomplete. What happens, Woolf asks, if you don't—or won't—stop? Charlie Chaplin's well-known late critique of film's advances and automatized motion in *Modern Times* (1936) is a send-up of both speed and robotic, mechanical motion that casts back to film's beginnings, which exchange motion for stillness as the butt of the joke.⁵⁶ Although Chaplin sounds a somber note in his film, earlier films that expressly avoid stillness celebrate their very preposterousness, linking the medium to speed's hedonism by way of its insistence on motion. And yet, at the center of these films remains the problem of stopping and of dealing with an object or person that gets in the way.

Consider, for example, Percy Stowe's film, *How to Stop a Motor Car* (1902), which is a trick film that relies on a sight gag for its refusal of stillness. In this film, a policeman, who initially tries to stop a car using hand signals, is instead run over by the car, which refuses to stop. The policeman's body is rendered piecemeal by the collision but then, miraculously, reassembles itself.⁵⁷ A second policeman arrives on the scene and subsequently demonstrates what is presumably the correct way to stop a car by bending over and having the oncoming automobile ricochet off his backside into a ditch. The humor behind this strange film reveals a stubborn attachment to motion at the expense of anything else, including the body of authority, which is scattered across the road or reduced to an animalistic rump. The film also speaks to the instability of **(p. 36)** a medium—automotive and cinematic—that will not (or cannot) stop on principle. The temporal instability at the heart of the car accident, which belies a tension between predictability and order—speaks to the instability of the new motion technology of the car, which, like early motion picture technology, was prone toward sudden and unexpected combustion and uncontrolled stoppage. Stoppage only happens as the punchline of a crude joke, when the car rebounds off the rear of the inspector, who is turned into a buffoon. The car does not stop for long; after a pause, it resumes its forward motion, while the officers pursue the vehicle in vain and on foot. The title of the film, *How to Stop a Motor Car*, is itself a joke, since the didactic purpose is a complete failure and the car does not, in fact, come to a proper stop.

How to Stop a Motor Car belongs to a group of early British films that depict car crashes as a means to send up stillness. Their unstoppable cars kick up clouds of dust and obscure the screen, explicitly denying the critical valence of stoppage or stillness beyond the trickery of a sight gag. Devoted singularly to movement, these short films presume to make a joke out of stillness by depicting cars that refuse to, cannot, or do not stop, turning the car crash and its threatening cessation of motion into an occasion for humor. Other titles, such as Cecil Hepworth's *How It Feels to Be Run Over* (1900) and *The Explosion of the Motor*

Car (1900), turn the most threatening thing about the new technology of the automobile—its potential to veer out of control, to collide with something in the way, to throw its passengers out—into a joke, emphasizing the medium's overdetermined commitment to motion at any cost. These films, which attempt to contain the accidental and the stilled with comedy and dispel the threat of collision, swerve away from representing stillness as something to be taken seriously, actively avoiding the stasis of a collision in favor of the confusion of motion's blur.⁵⁸ Both *How It Feels to Be Run Over* and *The Explosion of a Motor Car* contain the threat of the car crash and the sudden explosion, the residue of which (dust, smoke) becomes the blurred stuff of film. They enact a collapse of car, screen, and smoke, thus erasing their pictorial fields: *How It Feels to Be Run Over* overruns the pictorial space and overtakes the screen with motion itself; and *Explosion of a Motor Car* obliterates the central moving object, replacing the car with smoke.

Hepworth's single-shot film, *How It Feels to Be Run Over*, positions the camera itself as impediment or obstruction, which is at the heart of the film's perversity. This film acts as a counterpoint to Hepworth's own (and others', including Edison's) so-called phantom ride films, which were shot using cameras affixed to the front of moving trains. Adopting the weirdly disembodied point of view of the moving train itself, these (p.37) films foreshorten the view by surging into perspectival deep space as the foreground permanently gives way, rendered unstable and invisible by the ever-changing vantage. The vanishing point reverses its ontological position: instead of being the mathematical point of convergence in the distance where things disappear, it opens a field of visibility as the train approaches that is three-dimensional. The spectator/train tunnels into and through the pictorial frame, which disintegrates as the space of a moving landscape unfolds and disappears. These phantom rides were designed to impress on the viewer the sensation of train travel in front of the scrim of smoke and the train's steam so that speed might be felt, rather than seen symptomatically. *How It Feels to Be Run Over* asks its viewers to feel speed differently, however. Deploying an object that moves toward a camera (rather than the reverse), it asks the spectator to feel speed not as a plunge into and through the frame, but as an encounter with fixity—an encounter that will also threaten the terms of classical perspective and obliterate the pictorial frame, refusing to depict stillness itself.

In Hepworth's film, a fixed camera is positioned slightly to the left on an open country road, which curves back into the distance. Recalling the perspective and iconography of early train films, first we see a cloud of dust in the background, and then a horse-drawn buggy makes its way down the road, exiting to the right of the camera as it cuts a diagonal swath across the screen.⁵⁹ The cross-screen movement of the horse and buggy, an outdated mode of transport, establishes the depth of field of the shot and calls attention to the camera's position on the road itself.⁶⁰ However, both of these establishing elements—perspective and the

camera—will be negated by the final frames of the film. As the horse cart exits, the screen fills with blowing dust, which obscures the subsequent approach of a motor car that comes barreling down the road, creating its own voluminous clouds. A sometimes-visible bicycle travels along the left side of the car, unable to catch up, dancing in and out of the dust and at times blocked by the vehicle. Although it appears that the cyclist follows the horse and cart's path to the right, the deviant car, driven by Hepworth himself, swerves toward the camera and the wrong side of the road, its passengers waving their arms. The camera (establishing the point of view of the spectator) is overrun by the ever-growing image of the car, which presumably smashes into it.

According to the catalog description, "The car dashes full into the spectator, who sees 'stars' as the picture comes to an end."⁶¹ The end of the picture marks not only the close of the film, but also the end of the pictorial frame. The impact is not recorded—instead of a crash, the screen **(p.38)** goes black and the handwritten words flash quickly, "??!!!! Oh! Mother *will* be pleased", in a series of staccato frames that flicker as the spectator's "stars." The film thus averts any actual representation of the accident or stillness—the camera is not knocked over (at least not so that we see it), and the car does not end up in a ditch (at least not that we know). Rather than depicting the obliterated camera or stopped automobile, the film effaces the entire pictorial frame and the event of the accident itself by going black, sacrificing the image in a simultaneous anticipation and deferral of the car's collision. In its place, Hepworth scratched words on the surface of the film's emulsion. The only visible damage or impact, then, occurs on the surface of the film, which nonetheless keeps moving through the apparatus. In avoiding stoppage, Hepworth drives his car out of the picture, straight into film's black depths that extend far beyond the limits of the frame. Hepworth's motor car, it seems, cannot but be drawn to the fixed camera, even veering off course to the wrong side of the road to ensure its impact on film. The camera only becomes an impediment after the car makes it so.⁶² How, then, do we understand this unseeable accident? The scale of the car, which comes too close to the fixed camera, contributes to the collapse of the pictorial frame, which cannot accommodate the rapidly expanding body of the impending car. The screen goes black once the motor car has entirely blocked out the background and filled the screen, flattening it until it is surface and foreground only. The filmic surface and the screen converge and become the medium for writing, literally restoring legibility and reestablishing surface at the site of the accident.

Karen Beckman understands the insertion of writing on the screen in Hepworth's film as evidence of "the incorporation of cinema into another medium" (an expanded cinema, in other words).⁶³ The collision, itself unrepresentable, is made visible as writing, exceeding the limitations of the original medium. In this reading, and throughout her book on car crashes in film, Beckman is interested in the ways cinematic car crashes construct an

argument for cinema's hybridity—its messiness and impurity—which is produced at the scene of the crash. She argues that the paradoxical binding of speed and its cessation inherent in the car crash is a particularly apt figure for conceiving of the paradox of motion pictures—that their appeal to essential movement and continuity is underpinned and, at times, undercut, by their (unwanted) inhabitation by photographic stillness and fragmentation. The car crash in early films represents the ways in which the cinema is conditioned by the interplay between stopped and moving images, which, Beckman argues, complicates any claim to modernist purity or medium specificity based on motion alone.

(p.39) My take on these films is slightly different, however, for just as film papers over its photogrammatic basis by way of the continuity of projection, so, too, these films work to obscure stillness altogether. We do not see film's frames, nor do we see the stopped car. Instead, the films push at the representational limits of motion by covering stillness in clouds of dust and puffs of smoke. In the case of *How It Feels to Be Run Over*, any perceived impurity of cinema is, in fact, expunged rather than produced by the accident, which reinforces the stuff of the medium of film—screen and celluloid (that is, smoke and mirrors)—alongside its capacity to represent motion. Put differently, the words on the screen do not hybridize film (as Beckman argues), but render it visible as film at a particular moment in the early history of the medium when such claims to specificity were part of its development. Film's affiliation with motion shored up the currency of the medium and its shared affiliation with other motion technologies such as the motor car and the railway. However, its specificity is not that it moves, but that it screens motion—it plunges into space. It is for this reason that smoke becomes such an important figure for both speed (locomotive and automotive) and film, which is, in essence, a smoke screen.

Clouds of smoke, which augur both the moving car and the medium of film, are also part of a visual vernacular of trick photography that speaks to early film's association with magic and optical illusion.⁶⁴ The puff of smoke, deployed in films by Georges Méliès, is a visual trope of trick photography and illusion, allowing for the sudden appearance and disappearance of objects and people. For instance, Cecil Hepworth's *The Explosion of a Motor Car* (1900) repurposes the smoke and steam associated with the fast-moving car to make the vehicle disappear altogether, like a magic trick. In this film, a car carrying two men and two women is driving down a road, heading diagonally toward the fixed camera in the foreground. Before reaching the camera, the car suddenly explodes in an enormous puff of smoke, obscuring one-third of the screen. As the smoke clears, a nearby policeman (played by Hepworth) goes over to inspect the debris. He pulls out a telescope and looks up, only to have the passengers' dismembered body parts rain from the sky onto his head. The policeman proceeds to collect the limbs and organize them in a neat pile. He takes some notes and walks off. The accident is presented as a joke and, in the shift from realism to visual trickery, the film elides the real possibility of a car crash, swapping the potential

of a stilled collision for the preposterous or magical explosion—a big puff of smoke that obliterates the car and turns body parts into rag doll limbs. Stillness, which reinforces the singularity of the material world, is exchanged for filmic ether—smoke and projection, the substitution trick and the exploding machine.⁶⁵

(p.40) Speed Limits

Instantaneous photography operates under the signs of arrest and the unexpected, defying the desire for movement that early film exemplifies and inverting cinema's refusal of the limits of stasis. To get at the stakes of this refusal, I am interested in how these films, which are expressly about not stopping, inform how we see instantaneous photography's static rupture. I turn to a well-known photograph by the French photographer Jacques Henri Lartigue, *Le Grand Prix A.C.F.* (Figure 1.1), taken in 1912, after the invention of film and after film's apprehension of motion had been visually established. Lartigue started taking photographs when he was a young boy and took many thousands of snapshots over the course of his life. He finds his place in the history of photography at the intersection of forms of stillness and motion, having compiled an extensive corpus of images throughout his childhood that explore the capability of the camera to depict bodies and objects in motion: walking, running, jumping, swimming, falling, skidding, sliding, splashing, cycling, go-cart racing, floating, flying, driving. Lartigue's amateurism circumscribes his (p.41) attempts as a child to purpose a medium of fixity for the expression of motion.⁶⁶

Similar to the train that defers its arrival in *Arrival of a Train* by continuing past the fixed camera, in *Le Grand Prix A.C.F.* a racecar speeds past the limits of the photographic frame, carrying on beyond the purview of the image. We see only the back end of the car, with its tilted wheels that betray the slower speed of the camera's shutter lagging behind their warp speed. To take this photograph, Lartigue turned his camera itself into a moving vehicle, panning from left to right, but not rapidly enough to capture the entire car. Crucially, the image of the speeding car is a study both of technologies of stillness and motion and of the exchange of clarity and blur, whose obscurity is located not in the moving car but on the

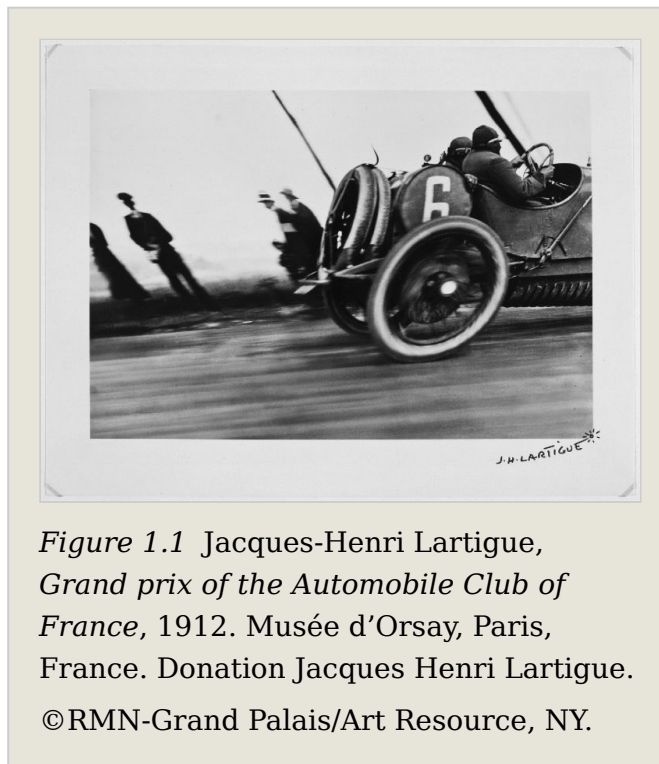


Figure 1.1 Jacques-Henri Lartigue, *Grand prix of the Automobile Club of France*, 1912. Musée d'Orsay, Paris, France. Donation Jacques Henri Lartigue. ©RMN-Grand Palais/Art Resource, NY.

edges of the vehicle and the image: the muddled spokes of the wheel; the horizontal striations on the road; the skewed, unfocused, and incomplete figures on the sidelines; the indistinct, washed out background. In this photograph, blur does not attach to motion per se—the racecar, its driver, and passenger are completely clear—but instead, blur figures motion’s periphery. Like the dust kicked up by the racecar, blur encases and abstracts most of the still elements of the photograph, whereas the moving elements, which are in focus, are nonetheless suspended out of sight. The faces of the driver and passenger are obscured both by their helmets, scarves, and goggles and by the position of the camera, which ends up behind its speeding subject. The car comprises only about half of the image. The rest is a still blur of foreground and background. In its representation of a moving vehicle, Lartigue’s image thus breaks open a schism between photographic clarity and blur, the visual condition that violates the legibility of the visual field and, under the rubric of instantaneous photography, enacts a series of exchanges between stillness and motion.

Writing about Lartigue’s photographs, Svetlana Boym claims that “he worked against the media”, wanting “photography to do what it cannot do, namely capture motion. Intentional technical failure makes the image at once nostalgic and poetic.”⁶⁷ Boym points to the blurriness and evasiveness of the images as signs of the failure of photography to properly still its subject. So, too, Enda Duffy notes that Lartigue’s photograph is a “limit-image”, which overtly signals the ends and edges of the photograph by way of the speeding car.⁶⁸ Although Duffy understands the car’s trajectory beyond the frame as indicative of the “inability of any image, even of a camera shot as bold and original as this one, to contain or represent a speeding car in full”, the photographic cut speaks as much to the photograph’s failure to fully contain its object as it does to the medium’s specific aestheticization in modernism of just such a limit or stop.⁶⁹ The cut marks **(p.42)** a necessary end point that signals the medium itself, calling attention to the photograph by way of its very finitude.⁷⁰ This is what makes Lartigue’s image nostalgic—its rendering of the end as a cutoff, as an arrest, which produces the conditions of longing that Boym describes.

The cut is not an essential feature of photography (Rosalind Krauss warns us against positing such a priori categories when thinking about medium specificity), but, as with Lartigue’s image, under certain circumstances and at a particular moment the fact of the crop or of limitation can indeed become specific to a medium.⁷¹ In the case of *Le Grand Prix A.C.F.*, the edge of the photograph formalizes its arrest and containment, its particular claim on stillness that interrupts and impedes continuity. In *The World Viewed*, Stanley Cavell also describes the photograph as a limited form that itself “comes to an end.”⁷² He suggests that an object that carries on beyond the edge of a photograph is not severed by the camera, but rather is “stopped live”, kept there or held in place in and by the photograph. This photographic stoppage—an arrest—distinguishes the finite photographic content from “the rest of the

world", which is cast aside or cropped out, rejected by the camera. Here, then, is the difference, almost crude in its simplicity, between film and photography: whereas the Lumière's train passes beyond the cinematic frame, Lartigue's car does not.

Although Lartigue's photograph does not depict a crashed car, it might not be entirely far-fetched to suggest that the vehicle does, in a sense, collide with the frame—a collision that effectively halts the car and cuts it off from continued forward motion and the world beyond the picture. Thus, the strangeness of photographic stoppage in Lartigue's photo, which relies on the accidental or contingent nature of the instantaneous photograph, is not so much that the car is severed by the frame, but that it is stopped at all and at the expense of a world that is not. This is, of course, the strangeness associated with instantaneous photography in general, which provides us frozen glimpses of the world that we would otherwise not see; instants that are shorn of a linear or narrative trajectory; instants whose arrest partitions time and space. Lartigue's image is meaningful precisely because it is halted, because the moving image is stilled. By drawing our attention to the crop and the edges of the picture, Lartigue turns finitude and limitation into a defining feature of the medium, and photography's stillness no longer registers as inadequacy in the face of speed. Instead, photography's limitation appears as necessary and distinct in the context of other modernist technologies of motion—the car and motion pictures.

Lartigue's picture of the car halted by the photograph offers a form of arrest that marks the terrain of photographic stillness in modernism and appears at a radical remove from motion pictures' insistence on **(p.43)** continuity. The atmospheric elusiveness of motion and the decomposition of the image remind us both that film is driven by the tautological effort to capture motion by way of motion and that photography is two-dimensional, marked as much by its capability to still and frame as by its limitations in so doing. Motion pictures summon photography's stillness as a limitation, casting it aside not only from motion pictures but also from human vision and experience.⁷³ Although speed was a quintessentially new modern sensation and cinema a new technology, it is stillness or stopping that is nonetheless marked as strange, unwanted, or unfamiliar.

Pausing the Frame

Perhaps the most well-known filmic commentary on early film's project to sideline stillness is Dziga Vertov's *Man with a Movie Camera* (1929). This film is central to Laura Mulvey's recent thinking about film's relationship to stillness in *Death 24x a Second*.⁷⁴ In a famous sequence from *Man with a Movie Camera*, discussed by Mulvey, Vertov directly couples moving film with a moving vehicle, granting film the ability to stop and start when the cameraman films a horse-drawn carriage by driving alongside and coordinating the relative speed of each by the actions of the camera. At the exact moment when the cantering horse

gets ahead of the camera, threatening to run faster than the car and leave the frame, the film freezes, holding the animal mid-stride and containing it with the stillness of a freeze-frame (Figure 1.2). Here, the film imitates the work of a photograph—to capture motion by stopping it. For a few minutes thereafter, the film operates as a slide show, moving from still to still, accumulating fragmented images of crowds, children, women. The film ultimately imparts order to these untethered images by revealing that the still image of a young girl exists as and in a sequence, on a film strip in the editing room, where the editor, Elizabeth Svilova, cuts and ultimately spins the film back to life. In her discussion of this sequence, Mulvey proposes that we understand the still frames in this scene “as photographs.”⁷⁵ Thus, she renders stillness analogical—we are assured that the freeze-frame, which is like a photograph, will indeed move again; we are assured that the mark of stillness is not a permanent stain. Mulvey writes that “the sequence seems to touch the point between the aesthetic of photography and the cinema.”⁷⁶ For Mulvey, that point is temporal and ontological—a point between past and present, symbolizing a moment of vivification, or animation, that also marks a transition between life and death. **(p.44)**

But the “point between the aesthetic of photography and the cinema” that so interests Mulvey also marks the shift between mediums in which one imitates the other. The still frames in *Man with a Movie Camera* are introduced as quotations: the aesthetic contact between photography and film relies on mimesis. The first quotation or film still of the cantering horse explicitly summons Eadweard Muybridge’s photographs of the horse in motion from 1878



Figure 1.2 Dziga Vertov, *Man with a Movie Camera*, 1929.

(Figure 1.3). These sequences, which were intended to reveal the nature of the horse’s paces, catch the horse mid-stride, in positions otherwise invisible to the human eye.⁷⁷ Muybridge abbreviates motion, cuts it off, and breaks it apart into incremental bits. The freeze-frame of the arrested horse in *Man with a Movie Camera* offers a nod, then, to the history of photography, sharing with it a claim on the medium’s revelatory potential, but, importantly, is not itself a photograph. Nor is the freeze-frame an image of a photograph. The cut of the image that stops continuity is performed in the editing room rather than by the still camera’s shutter. Thus, in Vertov’s film, prior to the image of the frozen horse, we see

repeated shots of the cameraman on the car, cranking the camera, moving the film through the apparatus—a cameraman who is not **(p.45)** a photographer, but, as the title of the film insists, a man with a movie camera. When the horse begins to trot once more, the camera is level with the animal's head as motion is recalibrated. The film's commitment to images of continuity and motion push stillness and the limits of the photographic frame to the side by deferring the cut, by running alongside the moving object rather than relinquishing it. Here, film imitates photographic stillness, but as a way to reassure us of its difference and distance.

The main narrative in this sequence of Vertov's film involves the editorial act of restoring motion and of putting the film to rights. Vertov sets up photography as a strawman, revealing stillness only so that it might be discarded in the editing room and ensuring that stoppage or stillness in film might only be superfluous or amusing. *Man with a Movie Camera*, like other early films, associates cinema with motion technologies to unframe the image, to widen the visual field

and include a beyond that carries on, out of sight. This beyond, for Vertov, implicates film's affiliation with atmosphere as a description of motion's spatial decomposition that is cast against an earth-bound stillness. As he explains in *Kino Eye*, his "mechanical eye" aims to express continuity and motion that does not stop, freed forever from "human immobility."⁷⁸ Opening spatial as well as temporal dimensions, he continues: "I draw near, then away from objects, I crawl under, I climb onto them." Pushing three-dimensionality into the atmosphere, Vertov ultimately claims the expanses of the sky as he imagines the **(p.46)** camera ascending with the airplane, plunging and soaring alongside aerial bodies in flight. In *Man with a Movie Camera*, the cameraman accordingly climbs rooftops and smokestacks, bringing the camera to the clouds. The film makes clouds central to its purpose, underwriting the medium with atmosphere: the initial image in the film is of a giant camera, which faces the screen and is surrounded by clouds. Behind the enormous camera, the cameraman climbs a hill and sets up his tripod as though in the air, coupling the movements of the camera with the passage of the clouds. Before the film continues by entering the interior space of the movie theater, the subsequent shots are again of clouds amassing and traversing space, which anticipate the many repeated images of



Figure 1.3 Eadweard Muybridge, *The Horse in Motion*, 1878. Library of Congress, Prints and Photographs Division, Washington, DC.

the sky, clouds, smoke, and steam that form a central part of the film's iconography.

The "human immobility" that Vertov says he wishes to escape seems to be more about being earthbound than about being still. To be completely still and to still something is not really human at all, but it is photographic or objective. As we have seen, the freeze-frames Vertov uses in the film are like instantaneous photographs: impossible images that would be invisible to the human eye without the intervening visual mechanics of stoppage. The film goes to great lengths not only to produce these still images, but also to dispense of them—to return mobility to the human apparatus of vision. "Human immobility", for Vertov, does not necessarily implicate photographic time, but instead describes a confined photographic space, which is limited to the planar dimensions of a single frame or grounded perspective. This is not to say that photography is tethered to the ground—the first aerial photograph was taken from an air balloon by Nadar in 1858—but that for Vertov, the limits of groundedness attaches to his notion of human immobility. By contrast, *inhuman* mobility (that is, the film's celebrated movement of trains, cars, planes, and film cameras) is three-dimensional and cinematic, dealing with the movement through diagonal depths, vertical heights, and horizontal lengths. Film becomes the movement of air, witnessed by clouds, smoke, and dust.

Blurring the Lines

The shift from photographic blurriness to cinematic blurring—from the two-dimensional stain of motion to the three-dimensional expanses of a puff of smoke—is anticipated in the motion studies of the French nineteenth-century scientist Etienne-Jules Marey, who spent much of his photographic career struggling with the visibility and representability of motion in two dimensions, inventing a series of devices and technologies (p.47) that push against the limits of photography. At the end of his career and life, he made a set of instantaneous photographs of lines of smoke, which evoke film's *sfumato*. These photographs of "smoke fillets", as Marey called them, taken between 1899 and 1902, depict otherwise invisible currents of air in the controlled space of a smoke machine to chart how the introduction of variously shaped objects alters the flow of air through a wind tunnel (Figure 1.4). What makes these photographs so resonant and so central to an understanding of how motion's blur was seen and figured in the early part of the twentieth century is their airy, smoky subject. These obscure images get at motion's invisibility or blurriness by representing a moving subject that is itself diffuse and particulate, cultivating a visual vocabulary of ephemerality that is central to film's embrace of motion over still and solid material forms.

The aim of Marey's smoke photographs was to determine how air might move around something that is in the way, how a still object might disturb or obstruct motion. Marey's late motion studies take seriously motion's (p.48) atmospheric and the solidity of an impediment, allowing that something might obstruct motion's trajectory. He accommodates stillness both as an object in the way and as a photographic method of representation, which runs counter to modernist ideologies of speed and motion that seek to refuse their opposite, at any cost. Although Marey's central concern throughout his career had to do with the scientific study of bodies in motion, these final instantaneous photographs deal, strangely enough, with motion's disturbance and the potential threat of its cessation. Stillness emerges as a necessity in Marey's final photographs, following a career that was devoted to the study of motion

as continuous. Given that these particular photographs were aimed at obtaining an understanding of aerodynamics to promote advances in aviation and the development of flight, the threat of unmitigated disturbance represented the real danger of a crash as a most radical kind of stilling. Marey's studies of air, read in the context of developments in instantaneous photography and early motion pictures, address how stillness in modernism is conceived of as an unwanted impediment—something that gets in the way of continuous motion—but also as a necessary interruption. Stillness accrues in an instant of illumination—a flash—that renders visible lines of smoke as moving currents.

Marey spent the bulk of his scientific career studying imperceptible aspects of motion—the quivering movements of a bird's wing, the traces of a beating heart, the rhythms of breath, the shape of a run, or the arc of a jump—obtained by way

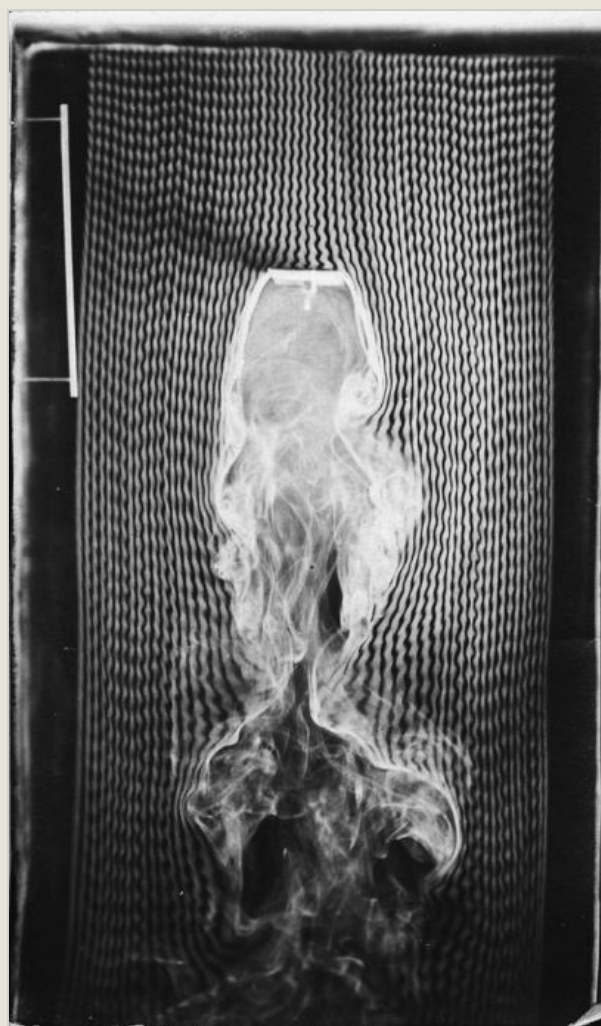


Figure 1.4 Etienne-Jules Marey, *Untitled* [Fumés-plan normal à la direction du courant], 1901. Cinémathèque Française, Paris.

of graphic inscription or chronophotography. In keeping with what Marey himself said about the mechanical objectivity obtained by his images, Joel Snyder has commented that the movements shown in Marey's images "fall outside the scope of human detection and accordingly, their inscriptions cannot be characterized as especially accurate visualizations of what might otherwise have been registered by an illustrator or scientist."⁷⁹ Snyder notes the logical impossibility of subjective mediation in Marey's photographs, which, created by "clockwork-driven inscribers fall outside the scope of human sensibility." The machinery of precision, designed to record and reveal the stages of motion, pursued a new articulation of sight that is beyond the scope of the subject and beyond the limits of human vision. Marey wrote, in *La méthode graphique*,

Not only are these instruments sometimes designed to replace the observer, and in such circumstances to carry out their role with an incontestable superiority, but they also have their own domain where nothing can replace them. When the eye ceases to see, the ear to hear, the touch to feel, or indeed with our senses give deceptive appearances, these instruments are like new senses of astonishing precision.⁸⁰

(p.49) Marey pits his instruments and their sensory capabilities against the instability or limits of the observing subject, setting up an unequal relationship between objectivity and subjectivity such that the machine, which can arrest motion, trumps human vision, which cannot. As Snyder points out, Marey goes on to consider mechanical objectivity on its own terms and at a remove from the subject entirely: "machines can be constitutive of their own field of investigation—one in which substitution is not at issue."⁸¹ This is an important articulation of the stakes of objectivity and stillness as they decenter the subject and allow for a visual order of precision that is abstract and unassimilated.

In his early work, Marey traced the motions of a body using various instruments designed to measure internal movement, such as the rhythm of the pulse, the cadence of animals in motion, or the arc of a bird's flapping wing. The body in motion was rigged with a series of wires and tubing that connected it with the stylus of a recording instrument. Marey's graphic method linked the body to the tracing machine. Later, Marey turned to photography, and the images for which he is best known are his chronophotographs, which concatenate multiple exposures within the frame in a series of overlapping stages taken from one single point of view. The images, which depend on the scale of the camera, are themselves quite small—the photographs miniaturize and reduce time and motion in the process of recording them. His chronophotographs depict motion as a successive series of blurred and overlapping positions, resulting, as Mary Ann Doane has noted, in an overabundance of detail and a lack of temporal precision.⁸² As is evident in the photographs of a running horse (remarkably different from Muybridge's photographs of the horse in motion), Marey's chronophotographs create a layered surface of temporal forms that press

against the constriction of the frame, composing an account of motion that is made up of blurred lines and overlapping bodies (Figures 1.5 and 1.6). This shadowiness or elision is key as Marey wanted to create an image of motion that would reduce the temporal gaps between the moving points on the body and that comported with continuity. Doane explains how the formal argument of these photographs indicates that “the truth of motion is imprecision of form and indecipherability.”⁸³ In other words, motion is blur. Both Doane and Marta Braun have noted that in the early decades of the twentieth century, the Italian Futurists elaborated on and celebrated this decomposition of form, which pulls Marey’s photographs away from their referent toward metaphors of motion, effacing the legibility of the image by way of blur.⁸⁴ The indecipherability of Marey’s images presses photographic representation toward allusion. They metaphorize or allegorize motion rather than depicting it. The goal is not to reconstitute motion, but to provide a visible form for ephemerality. **(p.50)**

The blurred abstractions in Marey’s chronophotographs are central for understanding his final photographs of smoke, which play on the iconography of his earlier images but are achieved through different methods. The smoke photographs, whose objective it was to visualize aerodynamic drag and determine how air might flow past the wing of a moving plane, provide the coda to Marey’s long-standing interest in aerodynamics that **(p.51)** began with a series of studies of the movement of insects’ and birds’ wings in flight. However, in these final images, his focus shifts from bodies in motion to the displacement of air—that is, from moving object to motion’s symptom—and the metaphorical blur of his earlier chronophotographs manifests subsequently as wiggly lines and puffs of smoke that twist around a still object in the way.



Figure 1.5 Etienne-Jules Marey, *Cheval au Galop*, 1886. Cinémathèque Française, Paris.



Figure 1.6 Etienne-Jules Marey, *Vole d'oiseaux*, 1887. Cinémathèque Française, Paris.

Thus, unlike his previous studies of flight, his smoke machines did not involve actual objects flying through the air, but air that moved against and over a solid form.

Marey's photographs of smoke share with his earlier motion experiments an epistemological goal of revealing something that is otherwise invisible for the sake of scientific knowledge (in this case, the invisible forces that construct the system of aerodynamics). But, unlike his earlier photographs, they do not deliver images that the human eye is incapable of seeing.⁸⁵ The filaments of smoke and their movements are discernable—you do not need a special lens to see the lines of smoke moving through the wind tunnel. What you do need, however, is stillness's mediation and a flash of illumination at the moment of the smoke's encounter with the impediment. These photographs of smoke expunge the durational surface of Marey's previous chronophotographs, replacing the blurred and prolonged temporality of chronophotography with smoke, which is a foil for both duration and motion. Marey's use of a flash was not new: in 1851, William Henry Fox Talbot photographed a piece of the London *Times* affixed to a spinning disk and illuminated by a spark of light, writing thereafter in a letter to Michael Faraday that "the image of the printed letters was just as sharp as if the disk had been motionless."⁸⁶ Talbot couples the success (here understood as clarity and precision) of the photograph by way of analogy, comparing the image favorably to its still counterpart (the paper itself) and branding motion as a contrivance. He predicts that with spark photography, "we shall be able to arrest and embody the most fugitive phenomena which present themselves to the eye."⁸⁷ Marey's engagement with air as a fugitive phenomenon inherits Talbot's promise of delivering still form to a moving object.

Between 1899 and 1901, Marey built four machines that allowed him to render air visible as lines of smoke, and in 1900 he presented his streamline technology to the Academy of Science in Paris: "Des mouvements de l'air lorsqu'il rencontre des surfaces de différentes formes." In her comprehensive work on Marey, Braun describes the *Machines à Fumée*, drawing from Marey's own description:

He built a ventilator to propel the smoke—made by turning tinder and cotton in a small furnace above—through twenty small tubes into the glass box. The (p.52) smoke passed from the tubes through a fine-mesh cloth and then over different shapes that Marey put inside the box. The camera, placed in front of the apparatus, recorded the movement of the smoke as it passed over the variously shaped planes. The pictures were made with a magnesium flash that lasted about 1/50 second, a burst of light that "surprises the fillets of smoke as they meander capriciously in the places where the eddies are formed."⁸⁸

The machine produced lines of smoke that were displaced by an object and photographed using a flash in a darkened room. The resulting images are completely dislocated—there is no context save for the squiggly smoke screen out of which billow abstract penumbra that symptomatize motion's disturbance or shock. The images document what happens when something gets in the way of movement, surprising it and shedding light on how momentum and entropy are each aesthetic categories produced by physical forces. And yet there is something strangely contradictory about the images, which has to do with their status as still instantaneous photographs. Marey choose to represent motion and smoke using the technology of instant photography rather than chronophotography or even cinema. Unlike his assistant, Georges Demenÿ, Marey was resistant to the cinema because, he claimed, it lacks the revelatory quality of photography, comporting instead with what we already can see. The images of smoke reveal the impact of stillness as a mode of resistance, as an obstruction to ephemerality. By employing still photography to apprehend smoke, Marey sets stillness in opposition to the cinematic threat of illusion and smoke's own dissolution over time: in a flash, the puffs of smoke are suspended; the undulations of the lines of smoke cease. Marey goes further, even, embracing stillness both as a method and as a function by introducing a still object into the visual field. The still object that causes the air's turbulence is an abstraction or shape that casts the filaments aside, turning them blurry, cloudy, and wispy in the wind tunnel. Although movement of the air is the subject of the photograph, not the still object, the shape of the air's movement depends on both the still impediment and the imposition of stillness and form. Marey's late photographs of smoke, which are subtended by stillness, nonetheless extend the metaphor of motion from his earlier images, pushing the analogy from blur to atmosphere.

Filming Air

The blurred filaments of smoke are residual—they make motion visible as an atmospheric abstraction, which is significant for early film. Although **(p.53)** motion pictures were celebrated at the outset for their ability to represent motion as it unfolded, the motif of smoke and blurriness nonetheless provided early films with motion's central metaphor, which, critically, film inherited from photography. In other words, the smudge of motion's airy blur carries over from one medium to the other. In his preface to *Theory of Film*, Siegfried Kracauer makes clear his notion that film "grows out of photography" and shares with its precursor an attachment to physical reality, charting a through-line from one technology to the next.⁸⁹ This reality, he writes, "includes many phenomena which would hardly be perceived if not for the motion picture camera's ability to catch them on the wing." Hinging the revelatory capacity of motion pictures to flight and atmosphere, Kracauer (possibly unwittingly) borrows from significant episodes in the history of photography that focus on wings: Talbot's microstudy of the wings of insects published in 1840 (Figure 1.7) and Marey's earliest

motion studies that involved transcriptions of the flight of birds and insects, achieved by tracing the movement of their wings in motion. Kracauer continues his (p.54) discussion of film's ephemerality in a similar vein, again evoking the air when he writes, "the cinema is conceivably animated by a desire to picture transient material life, life at its most ephemeral. . . . Significantly, the contemporaries of Lumière praised his films—the first ever to be made—for showing 'the ripple of the leaves stirred by the wind.'"⁹⁰ This unattributed phrase that attaches to cinema's invention—"the ripple of the leaves in the wind"—repeats throughout Kracauer's book as a leitmotif, recalling film's affiliation with the movements of air that feature in early films' engagement with catching the wind's visual symptoms—rustling leaves, clouds of smoke, steam emanating from a train.⁹¹

In *Signatures of the Visible*, Fredric Jameson also comments on film's fascination with moving leaves and their relationship to film and motion, borrowing from Kracauer the unattributed citation as well as his reference to Henry Cook and Gaetano Bonelli, whose photobioscope, a protocinematic device, promised images of landscapes "in which the trees bow to the whims of the wind, the leaves ripple and glitter in the rays of the sun."⁹² Jameson and Kracauer both cite D. W. Griffith's denunciation of Hollywood in 1947 for its lack of attention to the "beauty of moving wind in the trees" in his last interview.⁹³ The movement of the wind and the quivering leaves figure the restlessness of early film, its record of motion as both incidental and essential, and also mark its difference from photography, whose motionlessness cannot

apprehend the dimensions of these residual shifts in the air. The waving leaves in early film counterpose the necessarily absent element in one of Talbot's most

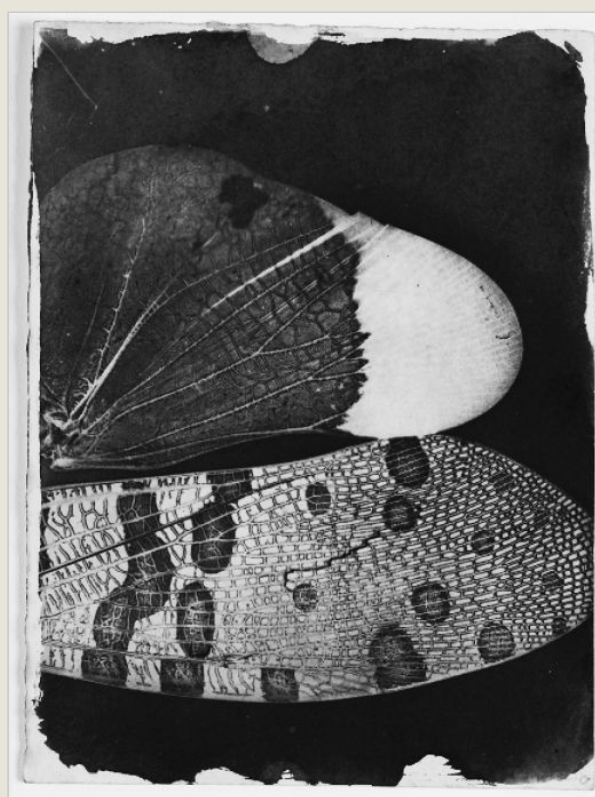


Figure 1.7 William Henry Fox Talbot, *Photomicrograph of Insect Wings*, 1840. Paper negative of a photomicrograph of insect wings, as seen in a solar microscope. SSPL/National Media Museum, Bradford, Great Britain/Art Resource, NY.

significant early photographs, *An Oak Tree in Winter* (1842), which is a picture of a massive old tree on Talbot's estate, whose extended branches are completely bare (Figure 1.8). This photograph was one of the first images to be fixed with a hyposulfite solution (developed by John Herschel), which stabilized the printed image. The fixity of the image is doubled by the permanence and solidity of its object, rooted to the earth and completely motionless. Leafless and silhouetted, the tree is stripped to its bones, enduring as a prehistoric cyclical element in the British landscape and existing outside of time and linear change. The moving leaves in early film, the ones that shift at the edges of the screen behind the head of the baby in the Lumières's *Le Repas de Béb * (1895), which caught the attention of M li s, are completely absent from Talbot's photograph. The photograph of the leafless tree is fundamentally still, while film instead makes something necessary of the incipient shifting movements of leaves in the breeze.

Yet by the time Marey is taking his smoke photographs, the movement of air, which we see evidenced at the edge of the frame in films like *Le Repas de B b *, becomes a photographic subject. It would be a mistake, then, to (p.55) separate film and photography on the basis of air. Rather, Marey's wind tunnels produce an important account of air's motion, stilled by the flash and shutter, that elaborate the terms of blurriness and define motion's visible trace that is taken up by film. Film's airiness is, then, essential to its understanding and its history, but its relationship to representing the ephemeral (or what Kracauer calls the "transient") is effectively photographic. Film adopts photography's blurred metaphors of motion that are established in Marey's late photographs and other still images of motion, which cast blur as the residue of speed—something that cannot be seen clearly and defies the legibility of the visual field to obscure stillness.



Figure 1.8 William Henry Fox Talbot, *An Oak Tree in Winter*, 1842–1843. Salted paper print, 19.4 × 16.6 cm. The J. Paul Getty Museum, Los Angeles.

The Ends of a Medium

Motion's ephemerality, which Gilles Deleuze describes as the continuity of movement, places an emphasis on constant motion as atmosphere. To (p.56) see the atmosphere, then, is to try to render the air visible as form—to turn movement into abstract image. My argument is that film turns movement into a medium of smoke, dust, and air. Still photography, in contrast, represents motion by obscuring the visual field through blurriness, which facilitates and formalizes the fleeting nature of motion. Film thus retains motion's blur not as a technical failure, but locates it in the air itself to become motion's medium. The obscurity writ into forms of motion suggests that it is something that cannot be properly seen, that can only be indexed rather than rendered per se in the form of an image.⁹⁴

The appeal of motion in modernism lies, in part, with its association with the fleeting and the immaterial—the passage of time—which is aptly registered in early films by moving bodies and moving technologies, like speeding trains and cars. These technologies of motion produce the airborne effects that themselves are cinematic. Smoke and wind, wafts of tinted air, suggest that the blurry visual effects of motion are residual or symptomatic. The visual effects of motion rely on dissolution, disrupting the coherence and legibility of photography's visual field, whose edges are smudged with the blurry residue of dust and smoke. These airborne particles become markers of the new medium of film, bearing the material trace of film's blurriness and representing its own transience. Motion itself is rendered visible in film through blur and dematerialization, marking speed by particles of dust and wafts of smoke. Although the shared terrain (indeed inseparability) between moving pictures and moving subjects in early film has been firmly established, what is at stake here is how film grapples with motion's invisibility and abstraction. I have argued that film ultimately tropes motion as unseeable, illegible, and blurry and makes this ephemerality central to the new medium. Indeed, motion is part of the atmosphere, more essence than object, which film tries to capture. Filmic motion thus manifests as a set of qualities—smokiness, blur, dissolution—that attach to the immaterial (that is, movement) to render it a visible abstraction. The conclusion that I am pointing to is, perhaps, tautological: in film's imagining of motion, motion looks like film. But what I mean more precisely is that screened motion looks like qualities of filminess—blur, smoke, dust—visual tropes of ephemerality that are, in fact, shared with still photography and that make it difficult to see clearly.

Notes:

(1.) In *Theory of Film*, Siegfried Kracauer emphasizes the partiality of film to motion, quoting the French avant-garde filmmaker, René Clair: "If there is an aesthetics of the cinema . . . it can be summarized in one word: 'movement.'" Siegfried Kracauer, *Theory of Film: The Redemption of Physical Reality* (Princeton, NJ: Princeton University Press, 1997), 34.

(2.) For well-known analyses of *Arrival of a Train at La Ciotat*, see Tom Gunning, "An Aesthetic of Astonishment: Early Film and the (In)Credulous Spectator", in *Viewing Positions: Ways of Seeing Film*, ed. Linda Williams (New Brunswick, NJ: Rutgers University Press, 1995), 114–33; and Karin Littau, "Arrival of a Train at La Ciotat", in *Film Analysis*, ed. Jeffrey Geiger and R. L. Rutsky (New York: W. W. Norton, 2005), 42–62.

(3.) Laura Marcus, *The Tenth Muse: Writing about Cinema in the Modernist Period* (New York: Oxford University Press), 19.

(4.) Lynne Kirby, *Parallel Tracks: The Railroad and Silent Cinema* (Durham, NC: Duke University Press, 1997). For more accounts of cinema and modernity's devotion to speed and its technologies, see Wolfgang Schivelbusch, *The Railway Journey* (Berkeley: University of California Press, 1977); Jonathan Crary, *Techniques of the Observer: On Vision and Modernity in the Nineteenth Century* (Cambridge, MA: MIT Press, 1992); Jonathan Crary, *Suspensions of Perception: Attention, Spectacle, and Modern Culture* (Cambridge, MA: MIT Press, 1999); Karen Beckman, *Crash: Cinema and the Politics of Speed and Stasis* (Durham, NC: Duke University Press, 2010); and Jeffrey Schnapp, "Crash (Speed as an Engine of Individuation)", *Modernism/modernity* 6, no. 1 (January 1999): 1–49.

(5.) Kirby, *Parallel Tracks*, 2, 3.

(6.) F.T. Marinetti, "The Founding and Manifesto of Futurism", in *Futurism: An Anthology*, ed. Lawrence Rainey, Christine Poggi, and Laura Wittman (New Haven: CT: Yale University Press, 2009), 51.

(7.) The stylized photographs of Edward J. Steichen, a founding member of Alfred Stieglitz's Photo-Secession Group, cultivate the misty, blurry effects of soft focus that are characteristic of Pictorialism. Larry Scharf, *Art and Photography* (New York: Penguin, 1974), 169.

(8.) O. Winter, "The Cinematograph", *New Review* (May 1896), reprinted in *Sight and Sound* 51, no. 4 (August 1982): 295.

(9.) See Yuri Tsivian, *Early Cinema in Russia and Its Cultural Reception*, trans. Alan Bodger (Chicago: University of Chicago Press, 1994), 146.

(10.) Rudolf Arnheim, *Film as Art* (Berkeley: University of California Press, 1958), 61.

(11.) Maxim Gorky, Review of the Lumière program at the Nizhni-Novgorod Fair, 1896, trans. Leda Swan, reprinted in *Kino: A History of the Russian and Soviet Film*, 3rd ed., ed. Jay Leyda (Princeton, NJ: Princeton University Press, 1983), 407–9.

- (12.) See James Cahill, "How It Feels to Be Run Over: Early Film Accidents", *Discourse* 30, no. 3 (Fall 2008): 296.
- (13.) Gorky, Review of the Lumière program, 407.
- (14.) Anonymous, "A Wonderful Invention: The Cinématograph of M. Lumière", *The Sketch: A Journal of Art and Actuality* 13 (March 18, 1896): 323. Also quoted in Tsivian, *Early Cinema in Russia*, 135.
- (15.) Quoted in Noël Burch, *Life to These Shadows*, trans. Ben Brewster (Berkeley: University of California Press, 1990), 26.
- (16.) Louis Delluc, "Photogénie", *Photogénie* (Paris: Editions de Brunoff, 1920). Delluc's "Photogénie" is not translated into English. This translation comes from the introduction to Laurent Guido and Olivier Lugon, eds., *Between Still and Moving Images* (Bloomington: Indiana University Press, 2012), 11.
- (17.) David Bordwell, *The Films of Carl Theodor Dreyer* (Berkeley: University of California Press, 1981), 42.
- (18.) Epstein, "On Certain Characteristics of *Photogénie*", trans. Tom Milne, in *Jean Epstein: Critical Essays and New Translations*, ed. Sarah Keller and Jason N. Paul (Amsterdam: Amsterdam University Press, 2012), 293.
- (19.) Clement Greenberg, the modernist voice of medium specificity, also refused to recognize photography as a full medium in its own right. In a review of Edward Weston's photographs from 1946, "The Camera's Glass Eye", Greenberg associates photography with literariness and sees it inadequate to the task of abstract (painterly) self-referentiality without sacrificing meaning. It cannot, in other words, be purely visual. He writes later in the *New York Review of Books*, "The art in photography is literary art before it is anything else: its triumphs and monuments are historical, anecdotal, reportorial, observational before they are purely pictorial." Clement Greenberg, "The Camera's Glass Eye", *The Nation*, March 9, 1946, 296; Clement Greenberg, "Four Photographers", *The New York Review of Books* 1, no. 11 (January 23, 1964).
- (20.) Epstein, "On Certain Characteristics of *Photogénie*", 293.
- (21.) Emmanuelle Toulet, *Birth of the Motion Picture* (New York: Henry Abrams, 1995), 130.
- (22.) André Bazin, "The Ontology of the Photographic Image", in *What Is Cinema?* vol. 1, trans. Hugh Gray (Berkeley: University of California Press, 1967), 14–15.
- (23.) Christian Metz, "Photography and Fetish", *October* 34 (Fall 1985): 85.

(24.) For more on stillness and film theory, see my essay on the contemporary artist Fiona Tan: Louise Hornby, "Stillness and the Anti-cinematic in the Work of Fiona Tan", *Grey Room* 41 (Fall 2010): 48–71.

(25.) Marinetti, "The Founding and Manifesto of Futurism", 51.

(26.) Schnapp reads the crash as the point after which Marinetti emerges as "the new metallic man with a pulsing light-bulb heart", the prophet of accelerated motion. See Schnapp, "Crash", 2.

(27.) Enda Duffy, *The Speed Handbook: Velocity, Pleasure, Modernism* (Chapel Hill, NC: Duke University Press, 2010), 3.

(28.) Duffy, *The Speed Handbook*, 4.

(29.) James Joyce, "After the Race", *Dubliners* (New York: W. W. Norton, 2006). Joyce interviewed the French driver, Henri Fournier, for *The Irish Times* prior to the race. Much of the interview involves determining what the average speed will be. Reprinted in the Norton edition, 222–24.

(30.) F. T. Marinetti, "The New Religion-Morality of Speed", in *Futurism: An Anthology*, ed. Lawrence Rainey, Christine Poggi, and Laura Wittman (New Haven: CT: Yale University Press, 2009), 226.

(31.) Quoted in Stephen Kern, *The Culture of Time and Space 1880–1918* (Cambridge, MA: Harvard University Press, 1983), 113. Rolls sought to accelerate the familiar scale of speed, the pace of a walk, that was established initially by the Highways and Locomotives Act of 1878, which mandated that cars not exceed the pedestrian speed of four miles per hour. The law was enforced by a flag-bearer who walked ahead of the car. By 1896, the red flag laws had been repealed, and in 1904, despite Rolls's protests, a speed limit of twenty miles per hour was legislated. This speed limit was systematically ignored and the number of car accidents increased dramatically. (The pace of a walk is the speed set in the "Proteus" section of *Ulysses* and photographed by Eadweard Muybridge in his motion studies, as discussed later in Chapter 3. So, too, the *pedestrian* pace of literature is what Marinetti seeks to redress in "The Technical Manifesto of Futurist Literature.")

(32.) "Future of the Motor Car", an interview with the Hon. C. S. Rolls, *The Guardian*, February 26, 1907. Rolls was devoted to exploring the edges of speed, himself establishing a world land speed record of ninety-three miles per hour in Dublin in 1903 driving a thirty-horsepower Mors. He raced bicycles at university, became an avid aviator, and was killed in an air crash on July 12, 1910.

(33.) Georg Simmel, "The Metropolis and Mental Life", *The Sociology of Georg Simmel* (New York: Simon & Schuster, 1950), 413–14.

- (34.) Walter Benjamin, "On Some Motifs in Baudelaire", in *Illuminations* (New York: Schocken, 1969), 155–200.
- (35.) Paul Virilio, *The Original Accident* (Cambridge, MA: Polity, 2007), 10.
- (36.) Marinetti, "The Founding and Manifesto of Futurism", 50.
- (37.) Lutz Koepnick, *On Slowness: Toward an Aesthetic of the Contemporary* (New York: Columbia University Press, 2014), 24.
- (38.) The Italian is "Che noia!" and the original French, "Quel ennui!"
- (39.) The German word for boredom, *Langeweile*, literally, "long while", makes clearer the association of the sensation of boredom with extended time. See Elizabeth Goodstein, *Experience without Qualities: Boredom and Modernity* (Stanford, CA: Stanford University Press, 2004), 108.
- (40.) As Joseph Pennell notes, following the invention of the motor car and the motorcycle, there was a dramatic decline in interest in bicycles. Joseph Pennell, "Cycles and Motors in 1900", *Contemporary Review* 68 (January 1901): 98–108. He continues this theme in Joseph Pennell, "Motors and Cycles: The Transition Stage", *Contemporary Review* 81 (February 1902): 185–95.
- (41.) Patrice Petro describes boredom thus: "a time without event, when nothing happens", in "After Shock/Between Boredom and History", *Discourse* 16, no. 2, A Special Issue on Expanded Photography (Winter 1993–1994), 78.
- (42.) Marinetti, "The New Religion-Morality of Speed", 225.
- (43.) On the "sluggishness associated with boredom", see Sianne Ngai, *Ugly Feelings* (Cambridge, MA: Harvard University Press, 2009), 270.
- (44.) Marinetti, "The Founding and Manifesto of Futurism", 51.
- (45.) See Winfried Menninghaus, *Disgust* (Albany: SUNY University Press, 2003), 48; and Sianne Ngai, "Afterword: On Disgust", in *Ugly Feelings* (Cambridge, MA: Harvard University Press, 2004), 332–54.
- (46.) Photographic return and the prolonged look are examined in detail in the final chapter of this book in terms of the maternal gaze in the novels of Virginia Woolf.
- (47.) In discussing the newspaper's account of the accident, Jeffrey Schnapp writes that "the cyclist pedals on indifferently", which is nowhere in the description that I can tell. Indeed, indifference is on the side of the driver and not the cyclist. See Jeffrey Schnapp, "Crash", 6.

(48.) Virginia Woolf, "The 'Movie' Novel", *The Essays of Virginia Woolf*, vol. 2, 1912–1918 (London: Hogarth, 1987), 290–91. MacKenzie's novel was later turned into a film called *Sylvia Scarlett* (1935), directed by George Cukor and starring (a cross-dressing) Katherine Hepburn and Cary Grant.

(49.) For further discussion of this passage, see Marcus, *The Tenth Muse*, 104.

(50.) Woolf, "The 'Movie' Novel", 290.

(51.) W. T. Stead, "The Church's Picture Galleries: A Plea for Special Sunday Cinemas", *The Review of Reviews* 46, no. 275 (November 1912). Reprinted in Colin Harding and Simon Popple, eds., *In the Kingdom of Shadows: A Companion to Early Cinema* (London: Cygnus Arts, 1996), 86–87. Cited in Marcus, *The Tenth Muse*, 104.

(52.) Stead, "The Church's Picture Galleries", 86–87.

(53.) Marcus suggests that when writing "The 'Movie' Novel", Woolf is describing forms of early comedy and may have had little familiarity with other genres of film, which would account for why her take on cinema "was so at odds with the habitual criticisms of sentimentality." See Marcus, *The Tenth Muse*, 104.

(54.) Henri Bergson, *Laughter: An Essay on the Meaning of the Comic*, trans. Cloudesley Brereton and Fred Rothwell (Mineola, NY: Dover), 3.

(55.) On boredom and its relationship to distraction, see Patrice Petro, "After Shock/Between Boredom and History", in *Fugitive Images: From Photography to Video*, ed. Patrice Petro (Bloomington: Indiana University Press, 1995), 265–84, especially 274. On laughter and the desire to escape boredom, see Sara Crangle, *Prosaic Desires: Modernist Knowledge, Boredom, Laughter, and Anticipation* (Edinburgh: Edinburgh University Press, 2010), 110.

(56.) For a discussion of Chaplin's slowness, see Michael North, *Machine Age Comedy* (New York: Oxford University Press, 2009), 190.

(57.) Other films, like W. R. Booth's *An Extraordinary Cab Accident* (1903), also reanimate bodies after they have been run over. In this film, a pedestrian is run over by a cab; onlookers, including the cabdriver, gather to inspect the man's stilled body, who, at the end of the film having been presumably declared dead, jumps up and, together with his female companion, runs off, laughing.

(58.) Enda Duffy makes a similar claim about the general discourse of car crash in the twentieth century, which "represents a veering off from the actuality of the crash itself, a displacement, to different degrees, of its fearful reality." Duffy, *The Speed Handbook*, 201.

(59.) For a similar point, see Cahill, "How It Feels to Be Run Over", 301.

(60.) See also Beckman, *Crash*, 29.

(61.) Anonymous, "Hepwix" *Films for the Cinematograph, Edison Gauge* (London: Hepworth, 1903), 23. Cited in Cahill, "How It Feels to Be Run Over", 300.

(62.) Whereas Michael Brooke reads this particular film as an indictment of car culture and a warning about the arrogance of new motorists who expect impediments to move out of their way, the position of the camera, which is not in the line of the car at the start of the film, indicates otherwise. Since the camera is to the left of the moving vehicle at the start, its imposition is manufactured by the car. It is less that the motorist expects the camera to move out of the way than that the motorist heads straight for it. Michael Brooke, "How It Feels to Be Run Over", BFI Screenonline, <http://www.screenonline.org.uk/film/id/444674/>.

(63.) Beckman, *Crash*, 30.

(64.) For more on the trick film, see Tom Gunning, "'Primitive' Cinema: A Frame-up? Or The Trick's on Us", in *Early Cinema: Space, Frame, Narrative*, ed. Thomas Elsaesser (London: BFI, 1990), 95–103.

(65.) For a similar discussion of this film, see Beckman, *Crash*, 45.

(66.) For a remarkable analysis of Lartigue's "boyishness" in the context of a series of other modernist paragons, see Carol Mavor, *Reading Boyishly: Roland Barthes, J. M. Barrie, Jacques Henri Lartigue, Marcel Proust, and D. W. Winnicott* (Durham, NC: Duke University Press, 2007).

(67.) Svetlana Boym, *The Future of Nostalgia* (New York: Basic Books, 2008), 21.

(68.) Duffy, *The Speed Handbook*, 178.

(69.) Duffy, *The Speed Handbook*, 178.

(70.) In her essay on Stieglitz's expansive series of cloud photographs, Rosalind Krauss also proposes that "the cut, the crop" is crucial to the scale of the medium: "if photography duplicates the world, it does so only in pieces." Rosalind Krauss, "Stieglitz/Equivalents", *October* 11 (1979): 133.

(71.) Krauss, "Stieglitz/Equivalents", 140.

(72.) Stanley Cavell, *The World Viewed* (Cambridge, MA: Harvard University Press, 1979), 24. The full passage reads, "What happens in a photograph is that *it* comes to an end. A photograph is cropped, not necessarily by a paper cutter or by masking but by the camera itself. . . . The camera, being finite, crops a portion from an infinitely larger field; continuous portions of that field could be

included in the photograph in fact taken; in principle, it could all be taken. Hence objects in photographs that run past the edge do not feel cut; they are aimed at, shot, stopped live. When a photograph is cropped the rest of the world is cut *out*. The implied presence of the rest of the world, and its explicit rejection, are as essential in the experience of a photograph as what it explicitly presents.”

(73.) For an account of the difference between photography and human vision, see Joel Snyder and Neil Walsh Allen, “Photography, Vision, and Representation”, *Critical Inquiry* 2, no. 1 (Autumn 1975): 156.

(74.) Laura Mulvey, *Death 24x a Second: Stillness and the Moving Image* (London: Reaktion Books, 2006).

(75.) Mulvey, *Death 24x a Second*, 15.

(76.) Mulvey, *Death 24x a Second*, 15.

(77.) I discuss Muybridge’s motion experiments at length in Chapter 3, including his images of the horse in motion.

(78.) Dziga Vertov, *Kino-Eye: The Writing of Dziga Vertov*, ed. Annette Michelson, trans. Kevin O’Brien (Berkeley: University of California Press, 1984), 17.

(79.) Joel Snyder, “Visualization and Visibility”, in *Picturing Science, Producing Art*, ed. Caroline A. Jones and Peter Galison (New York: Routledge, 1998), 379–80.

(80.) Etienne-Jules Marey, *La Méthode Graphique dans les sciences expérimentales et particulièrement en physiologie et en médecine* (Paris, 1878), cited in Marta Braun, *Picturing Time: The Work of Etienne-Jules Marey (1830–1904)* (Chicago: University of Chicago Press, 1994), 40.

(81.) Snyder, “Visualization and Visibility”, 381.

(82.) Mary Ann Doane, *The Emergence of Cinematic Time* (Cambridge, MA: Harvard University Press, 2002), 49.

(83.) Doane, *The Emergence of Cinematic Time*, 84; Braun, *Picturing Time*, 300.

(84.) Bragaglia, however, distanced his practice of photodynamism from Marey’s chronophotography, which, in his estimation, did not blur the lines enough.

(85.) See Christoph Hoffman, “Superpositions: Ludwig Mach and Étienne-Jules Marey’s Studies in Streamline Photography”, *Studies in History and Philosophy of Science* 44, no. 1 (March 2013): 1–11.

- (86.) William Henry Fox Talbot, letter to Michael Faraday (June 16, 1851a). Document 6429 in the Correspondence of William Henry Fox Talbot Project. <http://foxtalbot.dmu.ac.uk/project/project.html>.
- (87.) Talbot, letter to Michael Faraday (June 16, 1851a).
- (88.) Braun, *Picturing Time*, 217–19.
- (89.) Siegfried Kracauer, *Theory of Film: The Redemption of Physical Reality* (Princeton, NJ: Princeton University Press, 1997), xlvii.
- (90.) Kracauer, *Theory of Film*, xlix.
- (91.) Georges Méliès is widely credited for noticing the leaves moving in the background following a screening of the Lumière's film *Le Repas de Bébé* (1895). See Doane, *The Emergence of Cinematic Time*, 177; and Nico Baumbach, "Nature Caught in the Act: On the Transformation of an Idea of Art in Early Cinema", in *Cinematicity in Media History*, ed. Jeffrey Geiger and Karen Littau (Edinburgh: Edinburgh University Press, 2013), 107.
- (92.) Fredric Jameson, *Signatures of the Visible* (New York: Routledge, 2013), 270.
- (93.) D. W. Griffith in Anthony Slide, ed. *D. W. Griffith Interviews* (Jackson: University Press of Mississippi, 2012), 217.
- (94.) In "Still Moving: The Revelation or Representation of Dance in Still Photography", Matthew Reason also argues that the "degree of blur in the photograph provides an indistinctness that is suggestive of something in motion" (43). Matthew Reason, "Still Moving: The Revelation or Representation of Dance in Still Photography", *Dance Research Journal* 35, no. 2–36, no. 1 (Winter 2003–Summer 2004): 43–67.